

MAROC3

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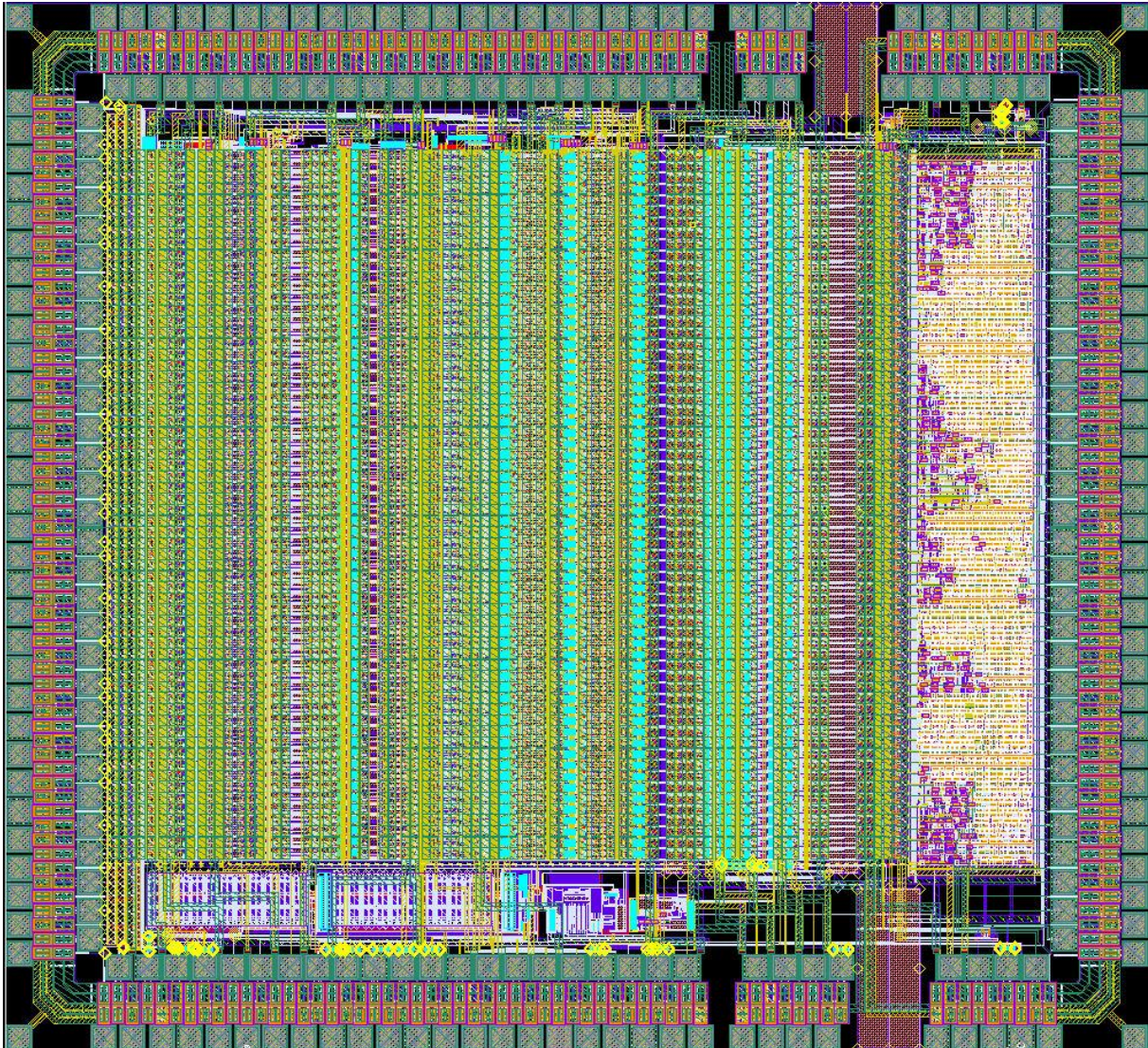
Abstract:

A new generation of circuit in AMS SiGe 0.35 mm technology has been designed for reading data from photomultiplier. One of these ASICs called MAROC (standing for Multi Anode ReadOut Chip) is an evolution of the 32 channels OPERA_ROC ASIC developed and installed on the OPERA experiment to auto-trigger and readout 64 channels Hamamatsu multi anode PMTs. MAROC is expected to discriminate the 64 channels PMT signals and produce 64 corresponding trigger outputs. The charge measurement is also available.

The requirements of MAROC were defined by the ones given to the electronic chain of the ATLAS luminometer for which this chip was designed at first. In order to have the best detection efficiency a 100% trigger rate for signal greater than 1/3 photoelectron is required. This corresponds to a charge of 50fC for a PMT functioning at a gain of 106 (High Voltage = 900 V). The noise is expected to be less than 2fC. The crosstalk between neighboring channels should be better than 1%. Additionally the charge measurement should be feasible up to a signal of 30 photoelectrons with a linearity of 2%.

The third version of this chip is made of:

- 64 low impedance preamplifier with a variable 8 bit gain for each channel. This variable gain allows reaching one of the requirements by compensating the PM gain dispersion up to a factor 4.
- 64 trigger outputs coming from either a fast shaper and his low offset discriminator in order to have 100% trigger efficiency rate for input signal close to 10fC or a bipolar fast shaper with lower gain followed by his discriminator allowing to trig for higher input charge without saturation. There is a switch to select between the first and the second trigger as the trigger output. The thresholds are set by two internal 10 bit-DACs.
- 64 variable slow shaper (50-150ns) followed by 2 Track and Hold to provide a multiplexed analog charge output from baseline up to 15pC.
- A digital version of this measurement is also furnished by a 12 bit ADC Wilkinson

MAROC die snapshot**Chip characteristics:**

Technology	: AMS SiGe 0.35μm
Chip Area	: 16mm ² (4mm *3.9mm)
Package	: CQFP240
Power consumption	: 220mW (→ 3.5mW/channel)
Power supply	: 0-3.5 V

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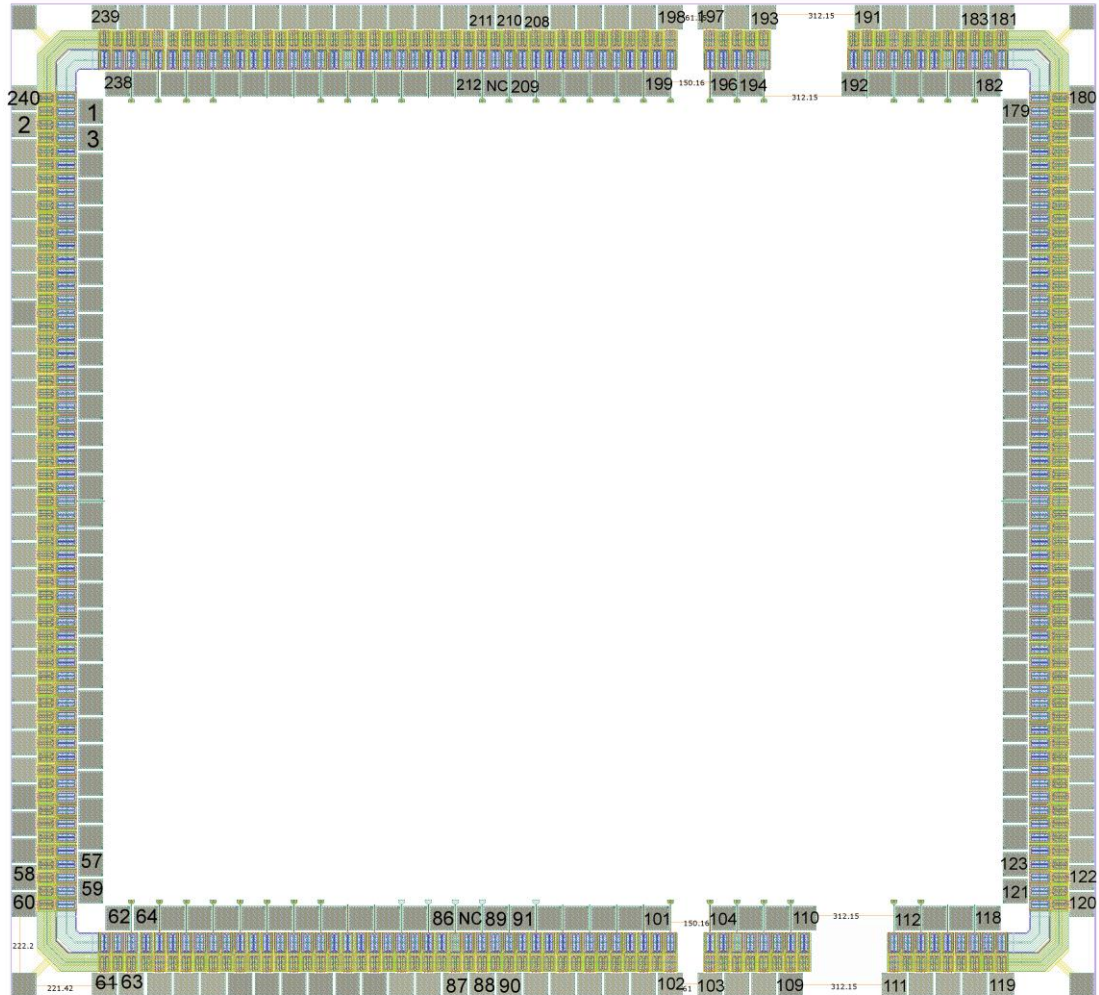
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3. MAROC3 chip Input/Output Pinout Table

MAROC3 is pin-pin compatible with MAROC2.

GND is the ground layer (0V) and AVDD and DVDD are power layers (3.5V) for the Asic.

The last column gives a cabling proposition.



MAROC3					
number	pin name	description	DC voltage	Bias (internal resistors and level value)	cabling
1	in<3>	detector	0.8V		Input
29	in<31>	detector	0.8V		Input
30	gnd_pa	Analogue (PreAmplifier) Ground	GND		GND
31	in<32>	detector	0.8V		Input
60	in<61>	detector	0.8V		Input
61	IN<62>	detector	0.8V		Input
62	IN<63>	detector	0.8V		Input

63	gnd_pa	Analogue (PreAmplifier) Ground	GND		GND
64	D_SC	Slow Control Register Input		TTL 3.3 from FPGA	Input
65	vssa	Inputs Bulk	GND		GND
66	RSTn_SC	Selected Register Reset	3.5V	TTL 3.3 from FPGA, Active Low	Input
67	vdd_pa	Analogue (PreAmplifier) Power	AVDD		AVDD
68	CK_SC	Slow Control Register Clock		TTL 3.3 from FPGA	Input
69	vgain_pa	Pre Amps bias voltage	2.5V	150Ω vdd_PA	NC
70	ibi_ss	Slow shaper input bias current	0.8V	125kΩ vdd_SS	NC
71	vcasc_pmos	Pre Amps bias voltage	1V	2,5kΩ vdd_PA 1kΩ gnd_PMO5 //	NC
72	ibo_ss	Slow shaper output bias current	0.8V	125kΩ vdd_SS	NC
73	gnd_nmos	Analogue (PreAmplifier) Ground	GND		GND
74	ibi_buf	Buffer bias current	0.8V	80kΩ vdd_BUF2	NC
75	vref_ss	Slow shaper reference voltage	1.1V	14kΩ vdd_BG // 11kΩ gnd_SS	NC
76	Qbuf_R	Multiplexeur Register Output		TTL 3.3 to FPGA	Output
77	gnd_w	ADC Ground	GND		GND
78	CK_R	Multiplexeur Register Clock		TTL 3.3 from FPGA	Input
79	Hold1	Hold Signal		default value=3,3V or 3.5V	Input
80	RSTb_R	Multiplexeur Register Reset	3.5V	TTL 3.3 from FPGA, Active Low	Input
81	Hold2			default value=3,3 or 3.5VV	Input
82	D_R	Hold Signal		TTL 3.3 from FPGA	Input
83	vdd_w	ADC power	AVDD		AVDD
84	ibo_dac	10-bit dual DAC output bias current	0,6V	100kΩ vdd_DAC	NC
85	gnd_dac	10-bit dual DAC ground	GND		GND
86	iref_dac	10-bit dual DAC bias current	0,46V	305kΩ vdd_DAC	Bias: Rx to avdd and Rx to gnd if needed or NC
87	vdd_dac	10-bit dual DAC power	AVDD		AVDD
88	ibi_dac			15kΩ vdd_DAC	
89	vref_dac	10-bit dual DAC reference voltage	2V	50kΩ vdd_BG //200kΩ gnd_DAC	NC
90	vbi_tz	FSU input bias current	2,6V	170kΩ gnd_FSU	Bias: Rx to avdd and Rx to gnd if needed or NC
91	vdd_fsu1	FSU power	AVDD		AVDD
92	G_diode	FSU bias voltage	1V	1kΩ gnd_FSU//2,5kW vdd_FSU1	NC
93	E_fsu	FSU ground	GND		GND
94	vbo_tz	FSU output bias current	0.96V	40kΩ vdd_FSU2	Bias: Rx to avdd and Rx to gnd if needed or NC
95	vcasc_fsu	FSU bias voltage	1V	2,5kΩ vdd_FSU1 10kΩ gnd_FSU //	NC
96	vslope	ADC Ramp bias voltage	800mV	154kΩ vdd_BG 46kΩ gnd_Wilk //	Rx to avdd and Rx to gnd if needed or NC
97	vref_fsu	FSU reference voltage	1V	1,5kΩ vdd_BG 1kΩ gnd_FSU //	NC
98	ramp	Ramp output	test	Scope waveform	Output: Pin test

98	gnd_wilk	ADC ground	GND		GND
99	vref_ramp	ADC ramp reference voltage	0.8V	126kΩ vdd_BG // 74kΩ gnd_Wilk	Rx to avdd and Rx to gnd if needed or NC
100	vdd_wilk	ADC power	AVDD		AVDD
101	ib_integ	ADC bias current		15kΩ gnd_Wilk	NC
102	vssa	Inputs Bulk	GND		GND
103	vssm	Inputs Bulk	GND		GND
104	vbi_discri	Discriminator input stage bias current	0.8V	100kΩ vdd_discri	NC
105	vdd_discri	Mixed (Discriminator) Power Supply	AVDD		AVDD
106	vbm_discri	Discriminator middle stage bias current	2.5V	100kΩ gnd_discri	NC
107	vdd_discriADC	ADC (Discriminator) Power Supply	AVDD		AVDD
108	vbo_discri	Discriminator output stage bias current	2.5V	100kΩ gnd_discri	Bias: Rx to avdd and Rx to gnd if needed or NC
109	gnd_discri	Analogue (Discriminator) Ground	GND		GND
110	PWR_ON	should be set to vdd value (3.5V)			Input
111	vssd	Digital part Bulk	GND		GND
112	gndd	Digital (LVDS receivers & Digital) Ground	GND		GND
113	vddd2	Digital (LVDS receivers & digital) Power	DVDD		DVDD
114	TransmitOn	ADC Active data readout		VH-VL logic to FPGA	Output
115	out_ADC	ADC serial data ouput		VH-VL logic to FPGA	Output
116	start_ADC	ADC start input		TTL 3.3 from FPGA	Input
117	vbi_discriADC	Rx to vdda and Rx to gnd if needed	0,8V		Rx to vdda and Rx to gnd if needed or NC
118	out<63>	Trigger output		VH-VL logic to FPGA	Output
119	out<62>	Trigger output			Output
120	out<61>	Trigger output			Output
121	out<60>	Trigger output			Output
150	vssd	Digital part Bulk	GND		GND
180	out<2>	Trigger output			Output
181	out<1>	Trigger output			Output
182	out<0>	Trigger output			Output
183	gndd	Digital (LVDS receivers & Digital) Ground	GND		GND
184	RST_ADC	ADC reset input		TLL 3.3 from FPGA	Input
185	vddd	Digital (LVDS receivers & digital) Power	DVDD		DVDD
186	CKb_40M	ADC: 40MHz Clock		LVDS from FPGA	Input
187	VL	Low value of the trigger voltage and ADC outputs (700mV,100nF to gnd)	700mV	analog input voltage: can be changed	Input: Low value of the trigger voltage (700mV,100nF to gnd) or the ground
188	CK_40M	ADC : 40MHz Clock		LVDS from FPGA	Input
189	vssd	Digital part Bulk	GND		GND

190	VH	High value of the trigger voltage and ADC outputs(1.5V,100nF to gnd)	1,5V	analog input voltage:can be changed	Input: High value of the trigger voltage (1.5V,100nF to gnd)
191	OR_0	OR of the first discriminators		TTL output	Pin test
192	OR_1	OR of the second discriminators		TTL output	Pin test
193	vdd_discriADC	ADC (Discriminator) Power Supply	AVDD		AVDD
194	vth1	10-bit dual DAC output 1	100nF to gnd	pin test: threshold second discri	Output: Pin test
195	gnd_discri	The same signal, only the name is changed	GND		GND
196	vth0	10-bit dual DAC output 0	100nF to gnd	pin test: threshold first discri	Output: Pin test
197	vssm	Bulk inputs	GND		GND
198	vss		GND		GND
199	Qbuf_SC	Slow control register output		TTL 3,3 to FPGA	Output
200	v_bg	internal bandgap (value 2,5V) ouput	2.5V	modification between MAROC2 (external bandgap 3,3V) and MAROC3 (internal bandgap 2,5V)	internal bandgap (value 2,5V): only pin test
201	out_fs	Fast Shapers Output	1V // 2V	test output (1V for FSU or 2V for FSB1)	Output pin test: a buffer should be added
202	gnd_fsu	FSU ground	GND		GND
203	vb_otafsu	FSU bias voltage	2.7V	500kΩ gnd_FSU	NC
204	vdd_fsu2	FSu power	AVDD		AVDD
205	vbo_fsb	FSB output bias current	2,3V	100kΩ gnd_FB1	NC
206	gnd_fsb1	FSB ground	GND		GND
207	vbi_fsb	FSB input bias current	2.5V	60kΩ gnd_FB1	NC
208	vref_fsb	FSB reference voltage	2V	500Ω vdd_BG // 2kΩ gnd_FB1	NC
209	ib_w	Track&Hold bias current	0.8V	250kΩ vdd_w	NC
210	vdd_fsb	FSB power	AVDD		AVDD
ib_otaq				15kΩ vdd_otaq	
211	gnd_fsb0	FSB ground	GND		GND
212	ib_sum	Preamplifier sum current bias	1V	5,6kΩ vdd_ss	NC
213	EN_otaq	Enable of the multiplexed analogue output	INPUT	TTL 3.3V from FPGA: default value should be 3.3V	Input
214	sum8	Sum ouput			NC
215	out_q	Multiplexed analogue output	1V	Charge ouput	Output pin test: a buffer should be added to see the waveform at scope
216	sum7	Sum output			NC
217	gnd_capa	Track &Hold ground	GND		GND
218	sum6	Sum output			NC
219	gnd_ss	Slow shaper ground	GND		GND
220	sum5	Sum output			NC
221	vdd_ss	Slow Shaper power	AVDD		AVDD
222	sum4	Sum output			NC
223	vdd_buf1	Buffer power	AVDD		AVDD
224	sum3	Sum output			NC

225	vdd_otaq		AVDD		AVDD
226	sum2	Sum output			NC
227	gnd_otaq	Multiplexed analogue output driver OTA ground	GND		GND
228	sum1	Sum output			NC
229	vcasc_nmos	Preamplifier voltage bias	1.5V	2kΩ vdd_NMOS //1,5kΩ gnd_NMOS	NC
230	vbi_pa	Rx to vdda and Rx to gnd if needed	0,65V	500kΩ vdd_pa	Rx to vdda and Rx to gnd if needed or NC
231	gnd_nmos	Preamplifier ground	GND		GND
232	NC				NC
233	vdd_pa	Preamplifier power	AVDD		AVDD
234	Ctest	input signal for internal capacitors		Test input	input signal for internal capacitors
235	vssi	Bulk inputs	GND		GND
236	vdd_pad	Inputs Pads Protection	AVDD		AVDD
237	gnd_pa	Analogue (PreAmplifier) Ground	GND		GND
238	in<0>	detector	0.8V		Input
239	in<1>	detector	0.8V		Input
240	in<2>	detector	0.8V		Input

power pin name	Pin number	current (mA)		power pin name	Pin number	current (mA)
vdd_w	83	2.4		vdd_FSU2	204	5.2
vdd_DAC	87	0.65 (dac max) to 1.92 (dac min)		vdd_FSB	210	8.3
vdd_FSU1	90	1,4		vdd_SS	221	8
vdd_wilk	100	4.6		vdd_buf1	223	0.6
vdd_discr1	105	3,6		vdd_OTAQ	225	5
vddd2	113	7.1		vdd_pa	233	5.4
vddd	185	0,8		vdd_pad	236	0
vdd_discrADC	193	7.6				

Consumption Current ~ 62mA

4. General description

The MAROC3 chip is a 64-channel input front end circuit to read out MAPMT outputs. It provides one shaped signal proportional to the input charge and 64 trigger outputs. Two outputs are implemented to make OR of the first and the second discriminator in order to perform a charge measurement with an auto trigger. Figure 1 gives a simplified schematic of the whole chip with its main features.

Each channel is made of a variable gain preamplifier ("Super Commun Base") with low input tunable impedance (50-100Ω), a low offset and a low bias current (5μA) in order to minimize the cross talk. This variable gain allows adapting the gain depending on the detector choice, up to a factor 4 to an accuracy of 1% with 8 bits. This gain tuning is also convenient to switch off a noisy channel. The amplified current feeds then two paths:

- A slow shaper path which consists in a CRRC² shaper with switchable capacitances to allow tuning peaking time and gain and 2 Track and Hold Widlar differential buffer one to measure the baseline and other one to obtain charge measurement. These T&H blocks store the charge in a 2pF capacitor and deliver a multiplexed charge measurement with a 5 MHz readout speed.
- Two possible fast shapers selected with cmd_fsb_fsu switch are followed by a one discriminator to deliver trigger outputs. Another fast shaper with less gain is implemented to trig for higher threshold.
- CK_mux and d1_d2 commands allow selecting the wanted discriminator outputs as the trigger output.

The different blocks are described in the next section.

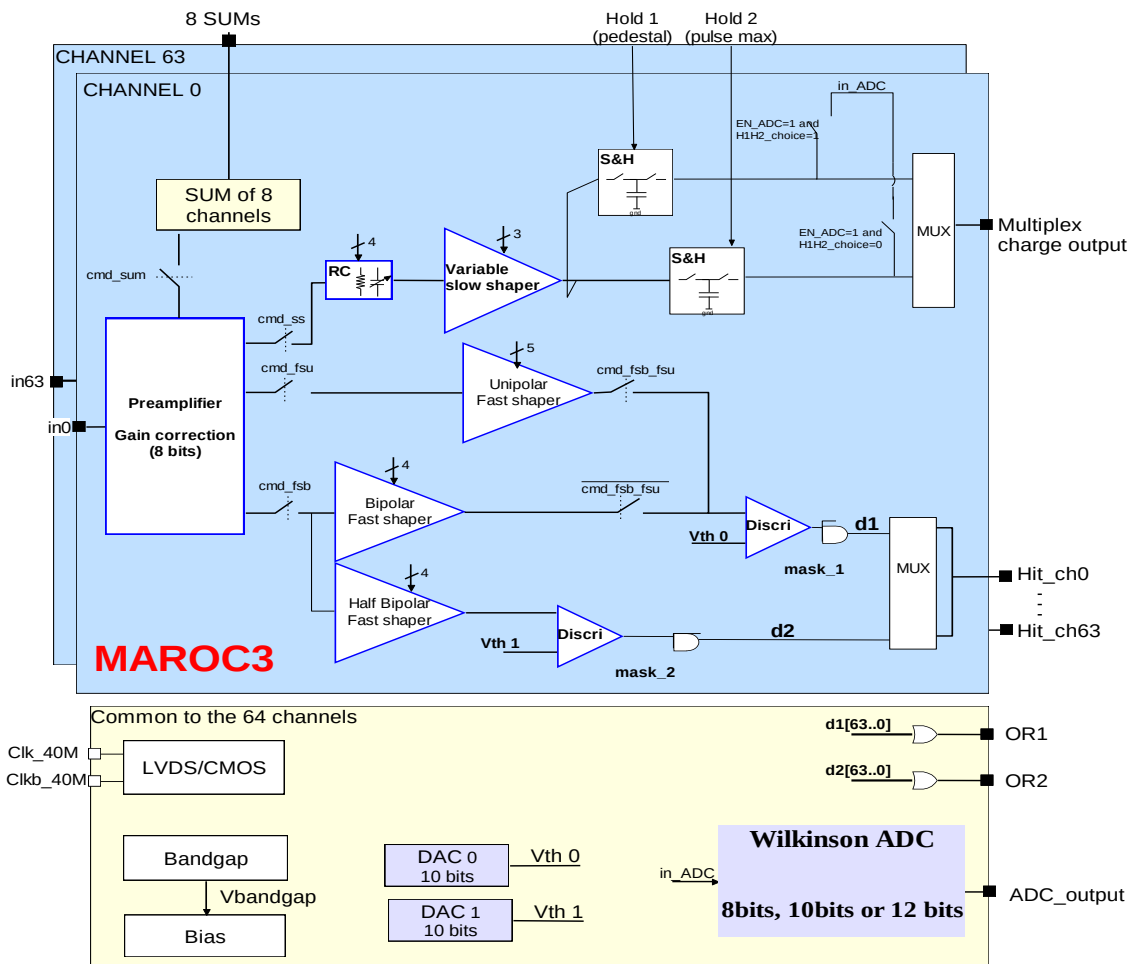


Figure 1: Block schematic of MAROC3 chip architecture

All the biases of each stage are common for every channels and have an internal default value referenced at a 2.5V internal bandgap. Each threshold is loaded by a 10-bit DAC and all switch commands are programmed through the Slow Control register. There are 829 Slow Control (SC) parameters which are loaded serially to control the chip.

Few blocks are power pulsed to decrease the power consumption: Bandgap, DAC OTA, Slow Shaper. A Read register is used to readout the charge output and the outputs of the selected fast shaper (Unipolar Fast Shaper "FSU", Bipolar Fast Shaper "FSB1", Half Gain Bipolar Fast Shaper "FSB2").

5. Preamplifier

5.1. Super Common Base

The preamplifier (PA) is a super common base to minimise the input impedance following by scaled mirror to set the gain for each channel. There are 8bits/channel to command these switches. These 8 bits correspond to a binary Gain tuneable between 0 and 255 and to an analogue gain varying from 0 up to 3.984 ($=1/64 + 1/32 \dots + 1/2 + 1 + 2$). The gain value 64 corresponds to the analogue unity gain and the gain of 0 allows inhibiting the signal at input.

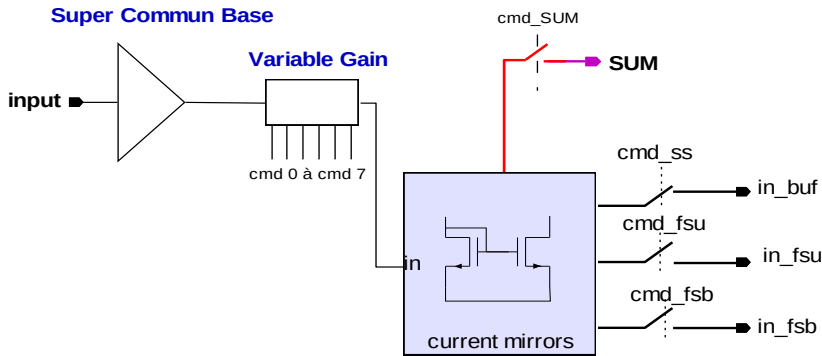


Figure 2: Block schematic

After the variable gain block, the current is copied to provide the sum outputs and to send or not signal at the slow shaper and fast shapers inputs (figure 2).

5.2. Sum outputs

5.2.1. Principe

We added the possibility to perform a sum up to 8 successive channel preamplifier signals.

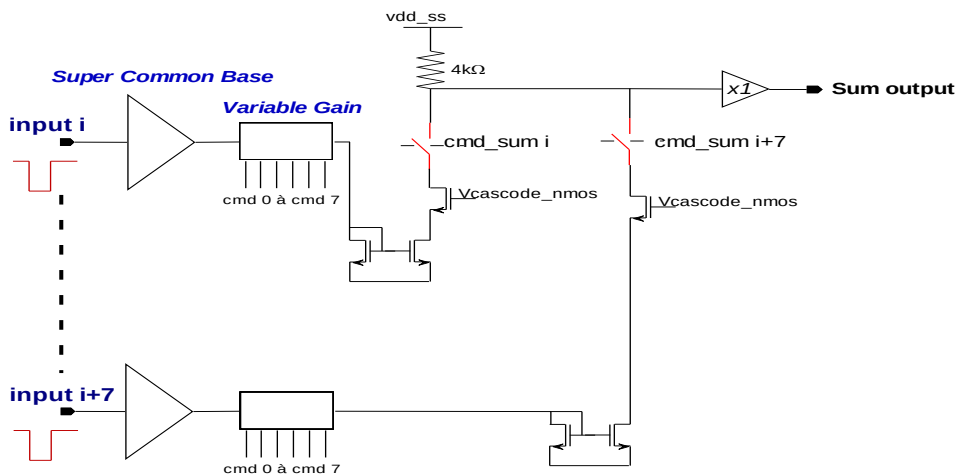


Figure 3: Sum schematic

5.2.2. Simulations

In this simulation, a same charge (100fC) is injected in all channels and the preamplifier gain is 1.

The DC voltage is proportional with the number of fibers used for the sum.

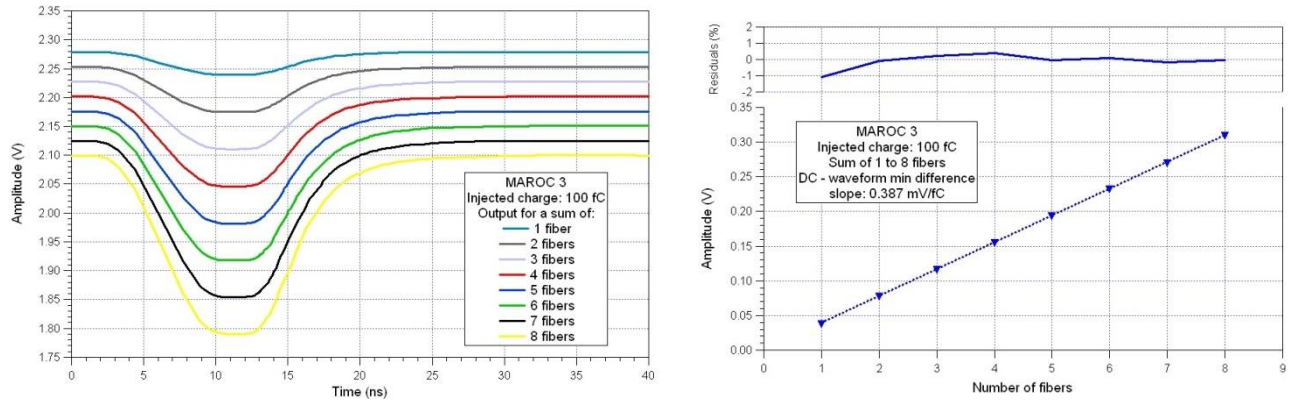


Figure 4: Sim: Waveform of the one sum output versus the number of fiber and linearity versus the number of fiber

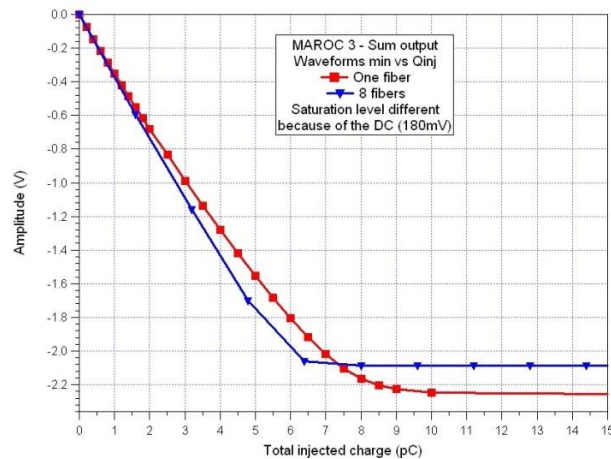


Figure 5: Sim: The sum linearity versus input charge for 1 and 8 fibers

6. Slow Channel

6.1. Charge measurement description

6.1.1. RC buffer and Slow Shaper

The preamplifier signal goes to the slow channel part thanks to a NMOS current mirror. The charge measurement will be performed through a RC buffer with variable capacitors to change the gain and a slow shaper to improve the signal/noise ratio at different rise time settings.

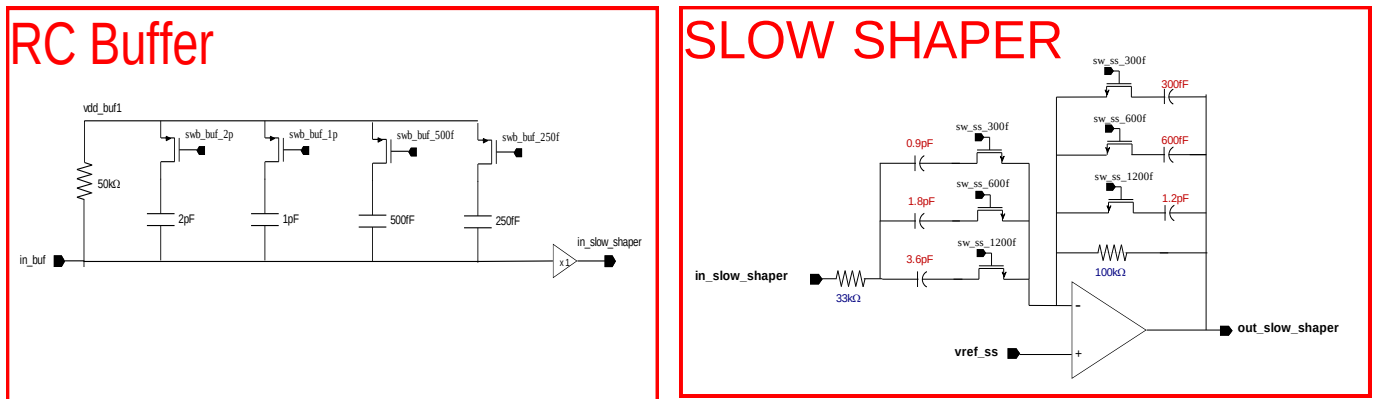


Figure 6: RC Buffer schematic and slow shaper schematic

Its gain and speed can be tuned changing the RC buffer network and SS feedback capacitor (SC parameters). The peaking time ($tp_{5-100\%}$) is expected to vary from 30 ns up to 150ns. ($tp_{5-100\%}=0.98\tau$). The SS is followed by 2 Track and Hold to store the baseline and maximum charge in a 2pF capacitor. The output can be read by selecting the channel using the Read register.

6.1.2. Measurements results

The analogue output has been studied as a function of different tuneable capacitors. There are 3 feedback capacitors on the slow shaper (C_{ss} varies from 0.3pF to 2.1pF) and 4 buffer capacitors (C_{buf} varies from 0pF to 3.75pF) common to all the channels, that can be switched on and off independently in order to vary the signal shape and its duration.

To reconstruct the waveform of the slow shaper, internal ADC conversions have been performed at different hold delays. This measure allows finding the best delay for the hold signal (hold value that corresponds to the signal maximum) for 8 capacitor configurations.

Converted data are plotted for different values of the hold (range 0-500, step 10n ns) in the following way: a pulse of 100 mV amplitude has been sent to a 10pF capacitor connected to channel 8.

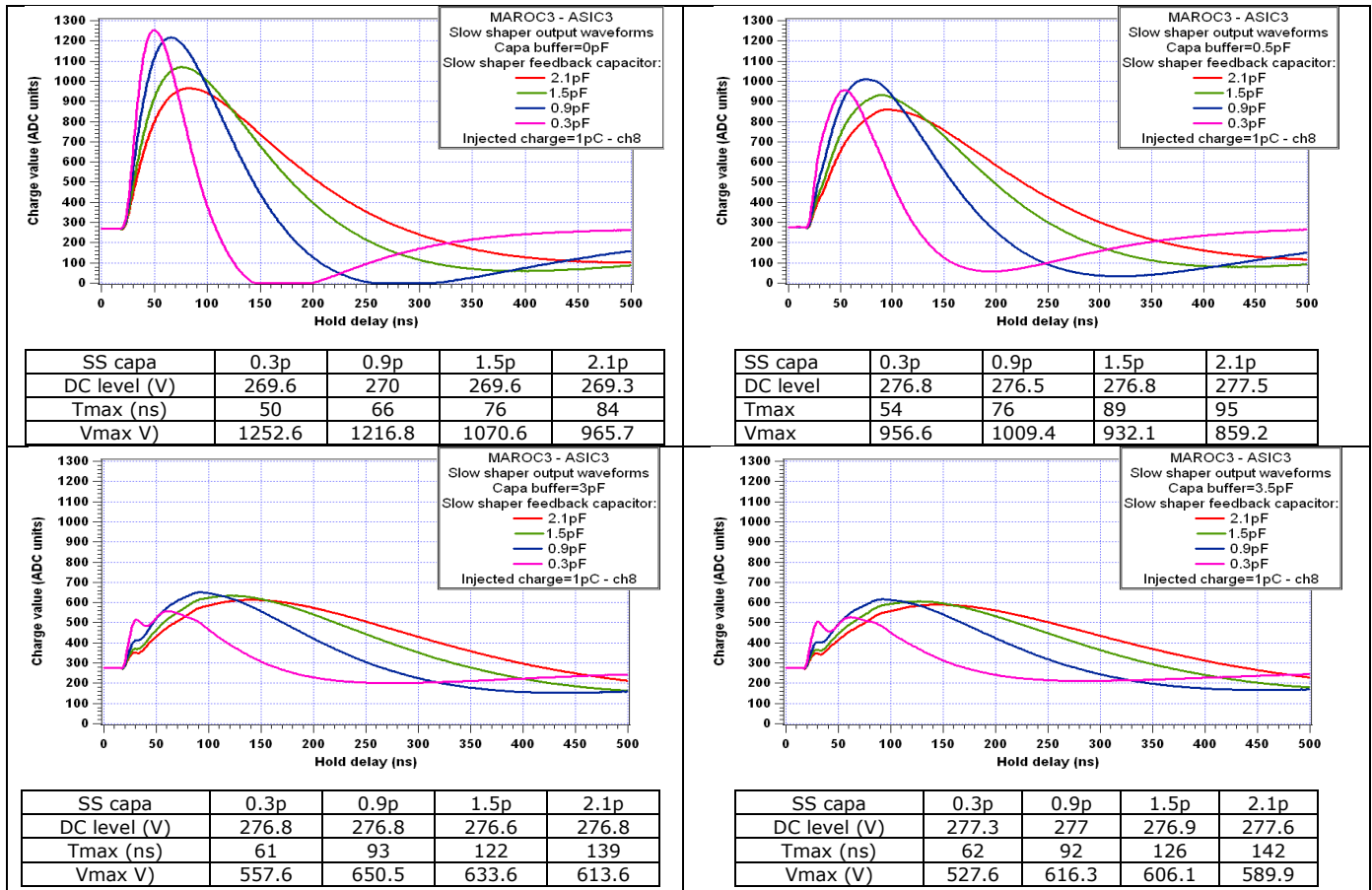


Figure 7: Measurement Slow shaper waveform versus buffer capacitances and feedback SS capacitances for the same input charge (1pC)

For few C_{buf} and C_{ss} combinations, we can see a coupling with the FSB1. But this problem doesn't disturb the hold position at the maximum.

6.2. Slow shaper linearity

A scan of the hold delay at few input voltage amplitude has been performed. The data converted as function of hold delay have been plotted for 3 shaping configurations.

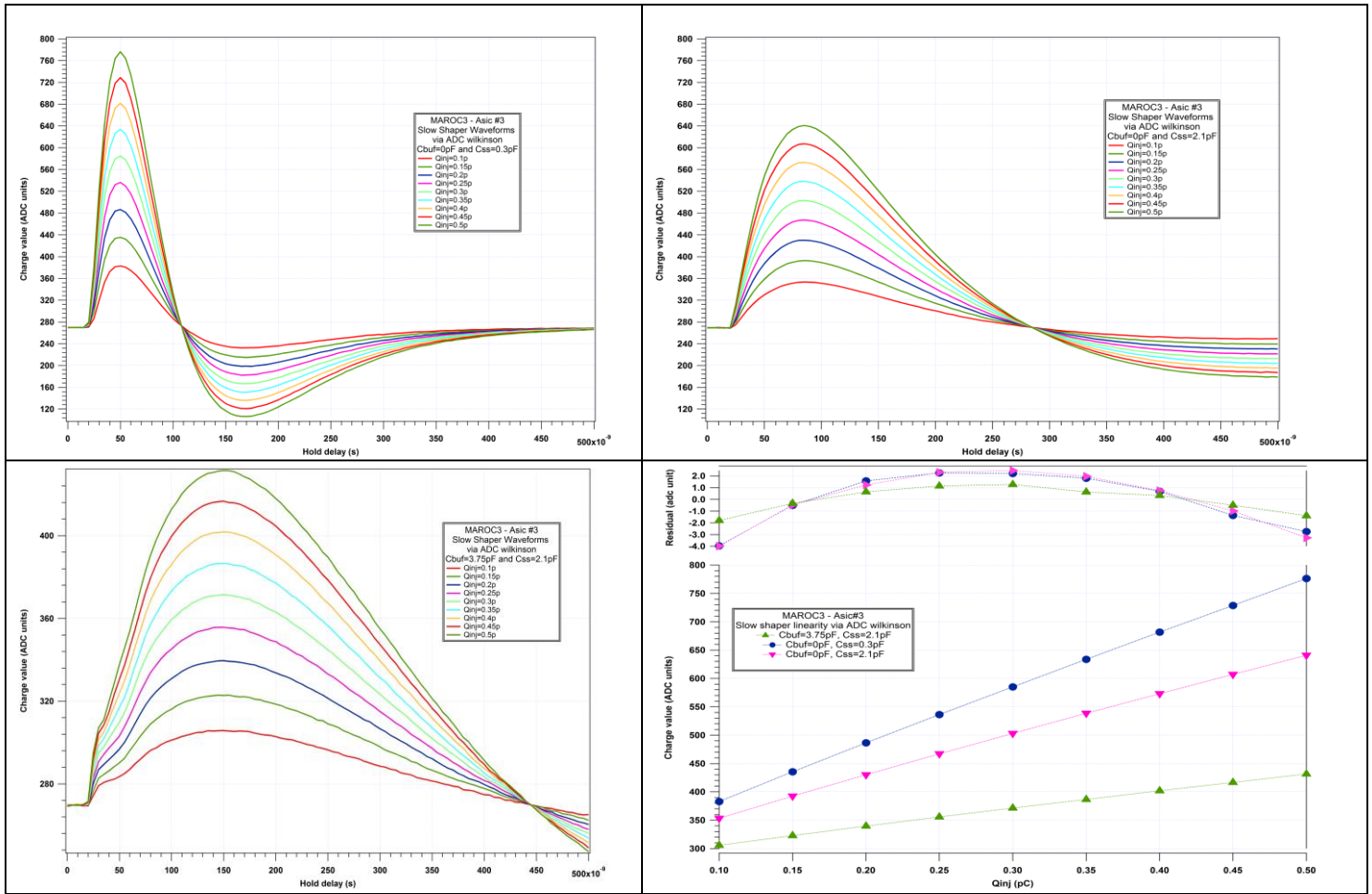


Figure 8: Measurement Slow shaper waveform versus hold delay at different input charge and Slow shaper linearity at different shaping time

config	0p+0.3p	0p+2.1p	3.75p+2.1p
Hold delay	50ns	85ns	150ns
gain	980.6 adc/pC	717.2 adc/pC	276.3 adc/pC
intercept	288 adc	286 adc	313a dc

6.3. Slow shaper linearity at different PA gain

With a fixed hold value of 100n (that means at the signal peak) and Cbuf=0pF and Css=2.1pF, a scan on the input voltage amplitude has been performed to test the linearity of the ASIC. The input charge has been varied from 0 to 30pC while 4 values for the preamplifier gain have been chosen (4 - 8 - 16 - 32 - 40).

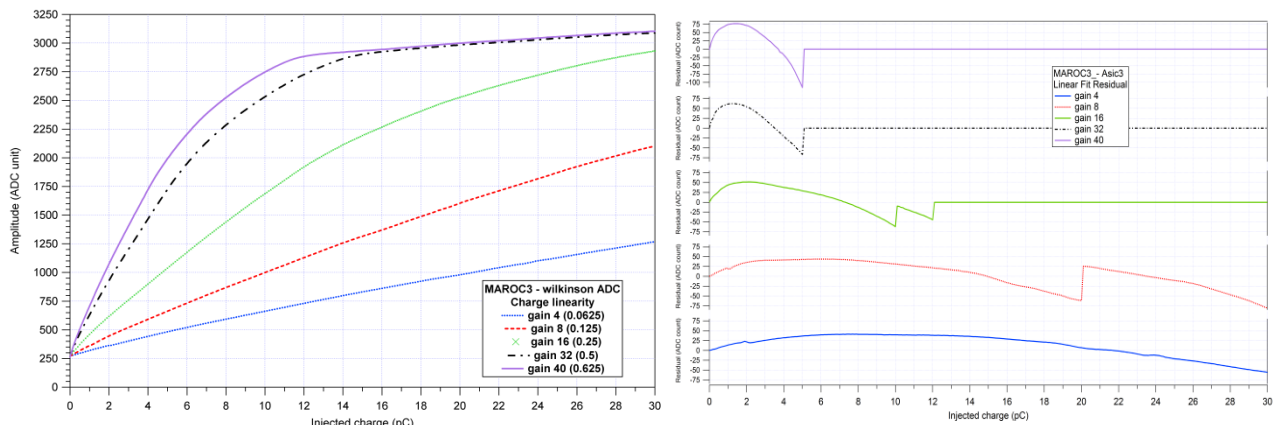
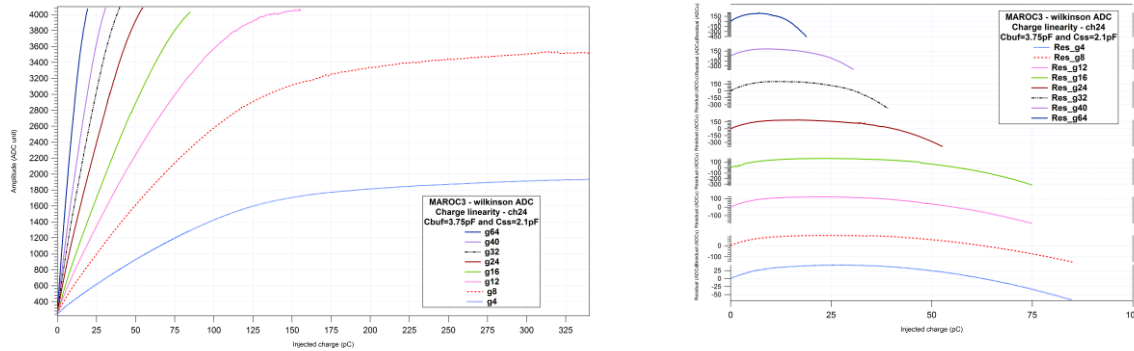


Figure 9: Charge Linearity at different gain

The table shows the gain of slow shaper and the limit of the fit to obtain a non linearity of about 2%.

Gain	4	8	16	32	40	64
Slope (adc count/pC)	35	70	148	305	383	636
Intercept (adc count)	271	271	269	265	262	271
Fit limit (pC)	30	20	10	5	4	3

The same measurement has been performed with the lowest shaping time: Cbuf=3.75pF and Ccs=2.1pF and hold delay=160ns.



The table shows the gain of slow shaper and the limit of the fit to obtain a non linearity of about 2%.

Gain	4	8	12	16	24	32	40	64
Slope (adc count/pC)	13	26	9	52	79	107	138	225
Intercept (adc count)	247	249	246	247	247	247	246	244
Fit limit (pC)	85	85	75	75	51	39.1	30.6	18.9
Data at fit limit	1292.5	2325.6	2991.7	3801.7	4026.7	4054.6	4073.5	4034.3

6.4. Crosstalk

To measure the crosstalk a large signal has been send in channel 8. The ratio between the adjacent channels and the channel 8 signal gives 0.25% crosstalk

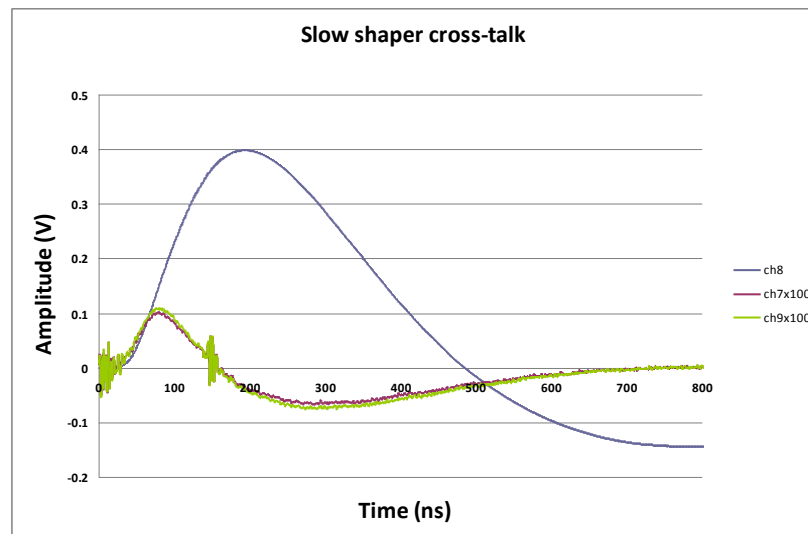
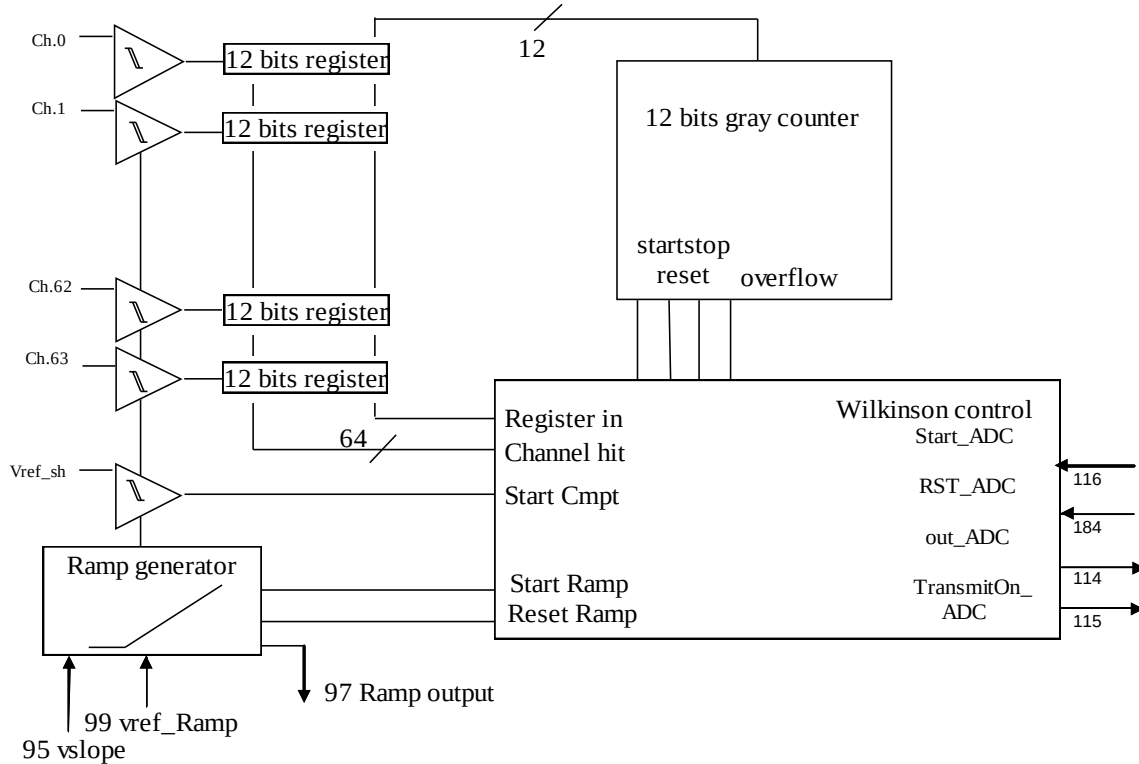


Figure 10: Slow shaper crosstalk

7. Wilkinson ADC

7.1. ADC description



The Analog to Digital conversion chosen is a Ramp ADC working at 40MHz. The voltage to digital conversion is performed by measuring the time between the start of a voltage ramp (or another position) and its crossing, detected by the comparator, of the voltage to be converted. The time measurement is achieved by a gray counter not started simultaneously with the ramp. Indeed the launching of the ramp is ordered by the start_ADC falling edge signal. The crossing between the ramp and the reference voltage of the slow shaper generates a trigger pulse which becomes the start of the gray counter. When a comparator triggers, its output is synchronized by the clock to memorize the state of the counter which will be the digital data.

The 12-bit gray counter and the ramp generator are shared between the channels. The ADC part replicated in each channel can be reduced to a discriminator and a memory used to copy and memorize the counter state when the discriminator triggers. So the power consumption and the area used can be very small even for high dynamic range.

The use of this kind of ADC is limited by its long conversion time. In fact, for an N bit conversion, it requires $2^N/Fck$, where Fck is the clock period of the counter. So for a 12-bit conversion, making use of a 40 MHz clock, 102.4µs are required.

On the test board, the RST_ADC is synchronised by the Hold1 pulse falling edge and stays at 0 during 25ns. The Start_ADC is synchronised by the rising edge of the RST_ADC pulse and have to stay at zero during 3 clock ticks.

The ramp begins at the Start_ADC falling edge and the readout of the data starts automatically at the end of the conversion. Data are synchronised with the rising edge of the TransmitOn_ADC.

Be careful: The TransmitOn_ADC and the out_ADC signals are either on pin 114 or 115 depending of the chip production run. The swapping can be easily done in the FPGA.

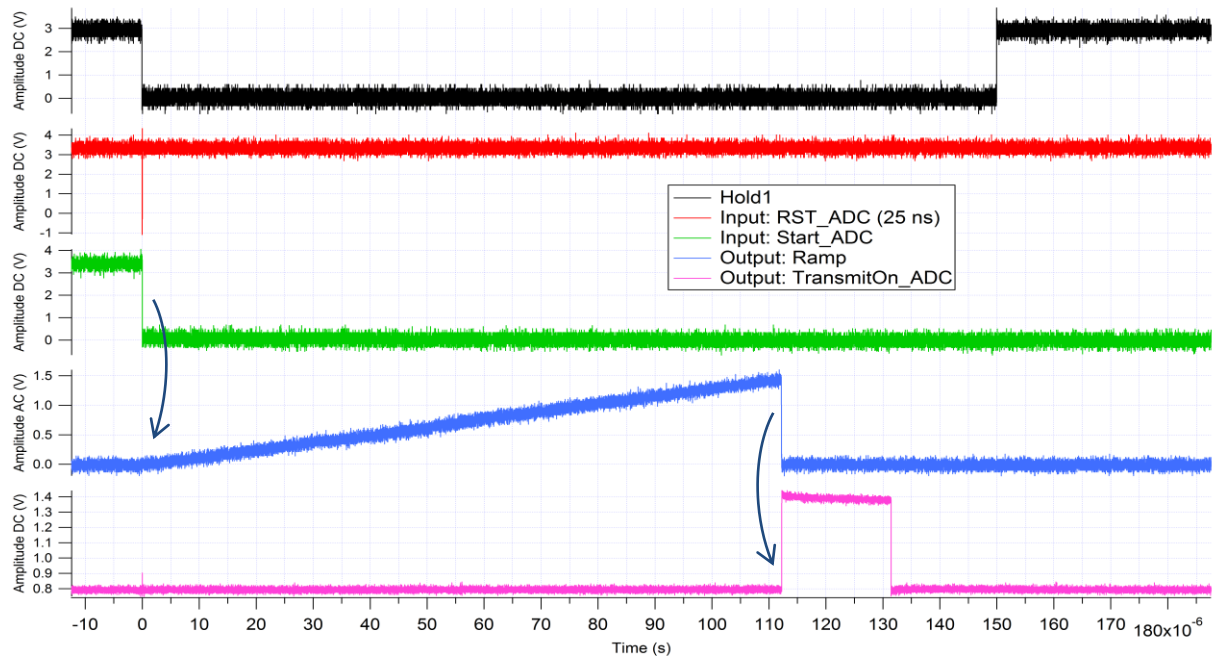


Figure 11: Wilkinson ADC

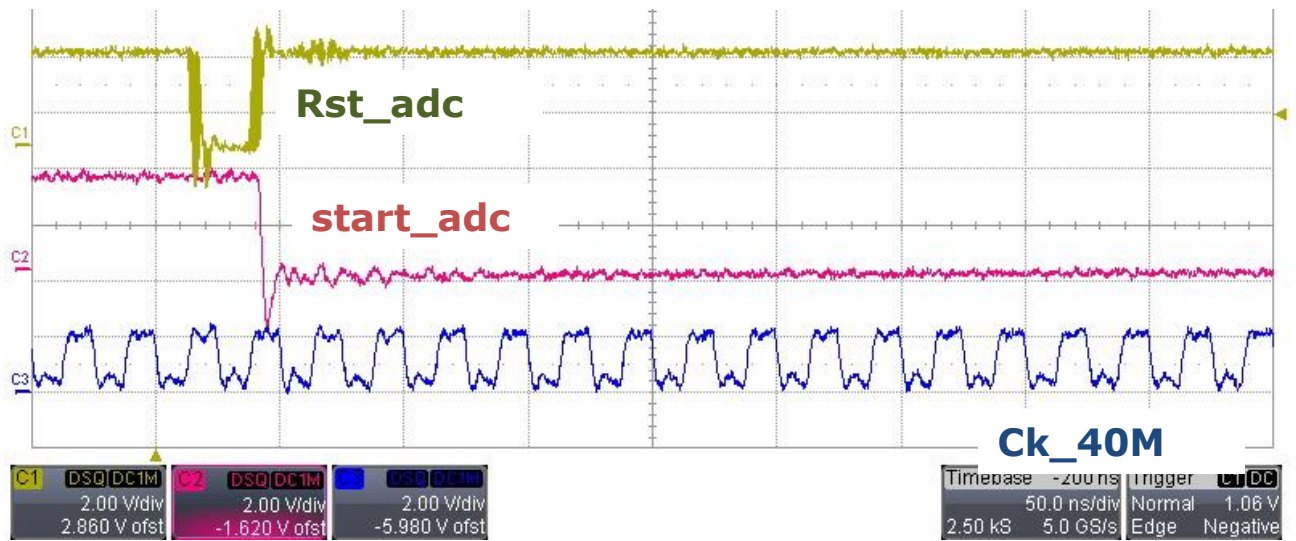


Figure 12: Wilkinson signal Zoom

Figure 11 shows that the ADC conversion only depends on RST_ADC, Start_ADC and 40Mhz clock. The first has to be low for one clock cycle (25 ns) and on its rising edge, Start_ADC goes low.

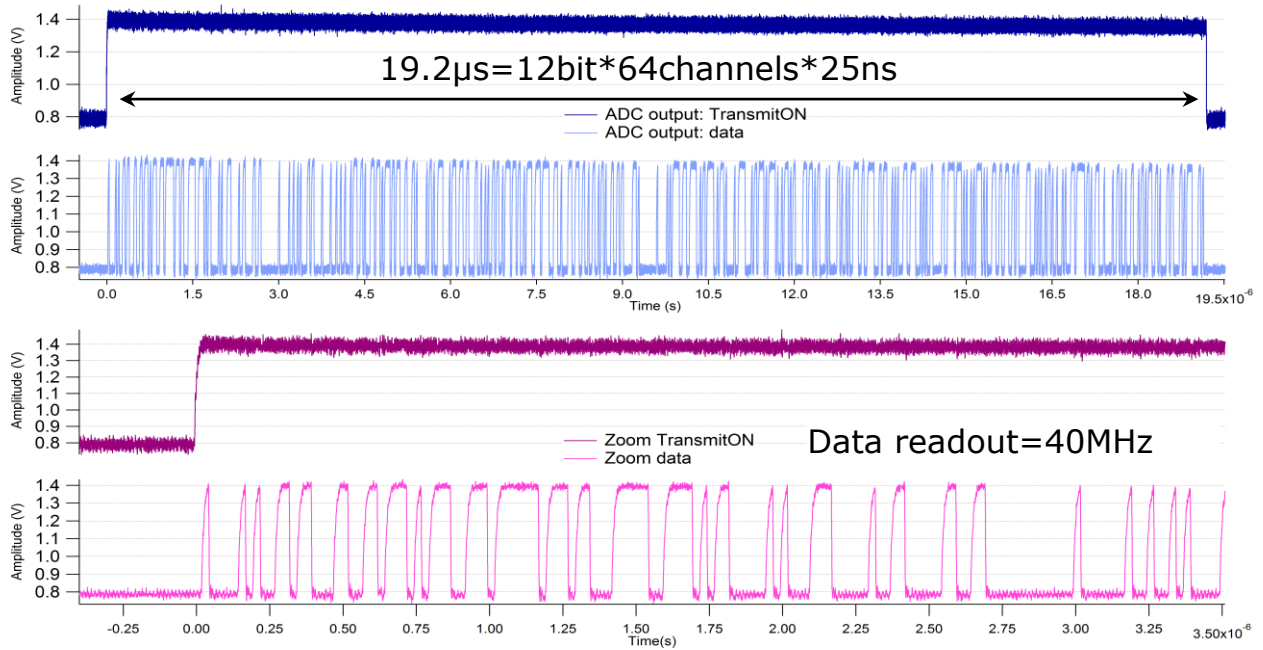


Figure 13: Wilkinson ADC digital outputs

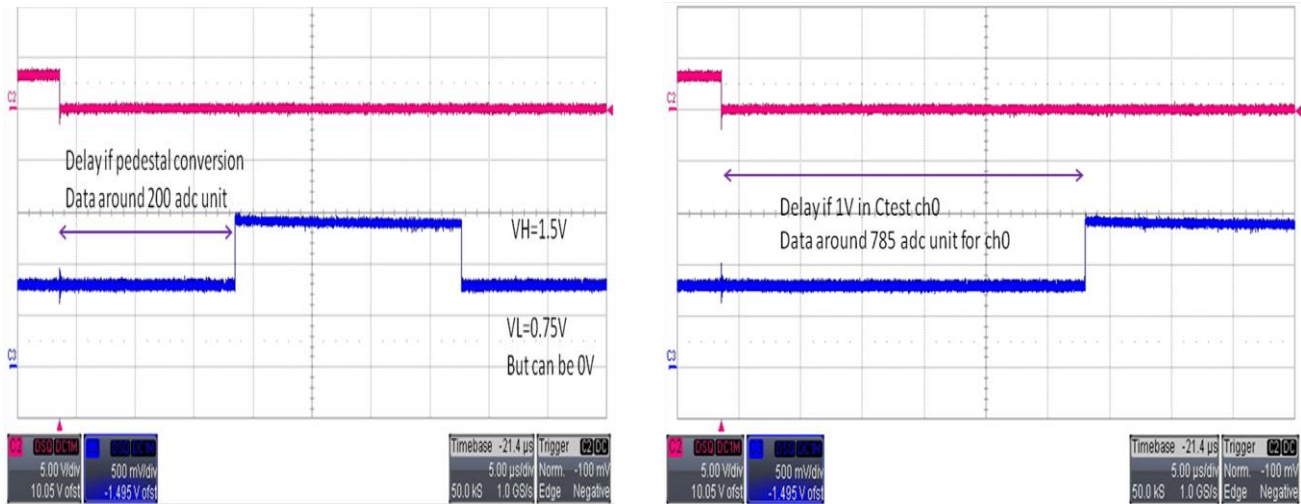


Figure 14: Delay between start_adc and transmitON

7.2. Wilkinson calibration and pedestals measurements

With this ASIC we don't have a dedicated input to calibrate the better reference and slope of the ramp. So we have to send a huge input charge to saturate the slow shaper and tune the ramp in order to have the highest data converted without to be at the over flow (means data equal 0). The saturation voltage of the slow shaper depends of the buffer capacitor and feedback capacitor settings.

The calibration has been performed with the highest shaping time: with $C_{buf}=3.75pF$ and $C_{ss}=2.1pF$ and with 10 different charge inputs. On the test board a **470kΩ** resistor has been added at the pin 95 to change the slope of the ramp and to be able to convert the saturation voltage of the slow shaper. ($V(\text{pin } 95)=0.584V$).

➤ 12 bits ramps

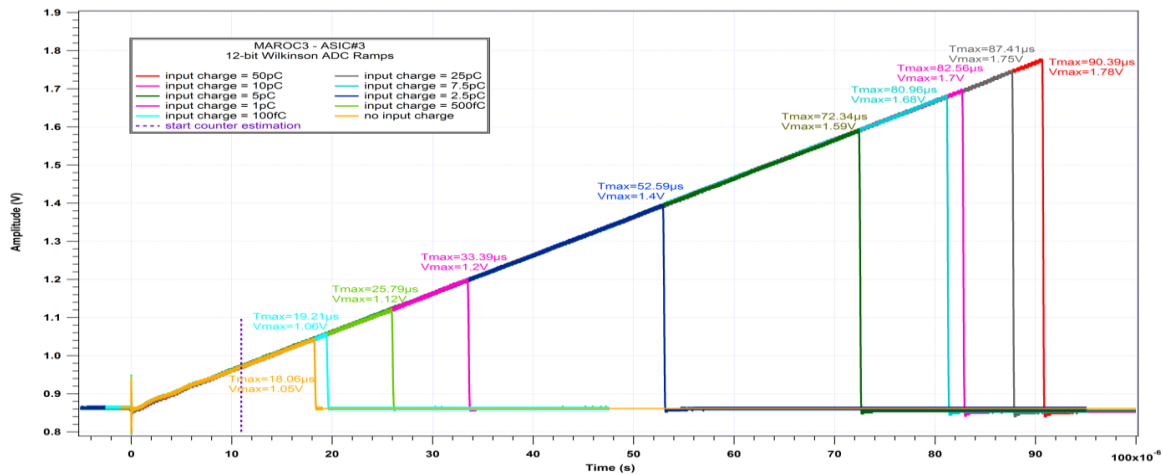


Figure 15: Ramp waveform for 12-bit conversions

Input charge (pC)	Tmax (μs)	Vmax (V)	Data (adc unit)	Rms (adc unit)	Start Counter (μs)	Voltage at start counter (V)	LSB (mV/adc unit)	slope (mV/μs)	intercept (V)
0	18.06	1.05	263	2	11.51	0.976	0.265	9.972	0.861
0.1	19.21	1.06	333.2	1.9	10.88	0.97	0.264	9.971	0.861
0.5	25.79	1.12	593.5	2.3	10.95	0.971	0.258	10.01	0.861
1	33.39	1.2	896.1	2.1	10.99	0.971	0.258	10.06	0.86
2.5	52.59	1.4	1674	2.2	10.75	0.969	0.255	10.08	0.86
5	72.34	1.59	2454	2	10.99	0.971	0.254	10.06	0.861
7.5	80.96	1.68	2803	2	10.88	0.97	0.254	10.09	0.86
10	82.56	1.7	2865	2.1	10.95	0.97	0.254	10.09	0.86
25	87.41	1.75	3063	1.8	10.83	0.969	0.254	10.09	0.859
50	90.39	1.78	3183	1.9	10.82	0.969	0.254	10.1	0.86
mean					10.95	0.97	0.257	10.05	0.861
rms					0.21	0.002	0.004	0.05	0.0007

Thanks to all ramp waveforms and the data converted the temporal position the start counter can be estimated at 10.95μs. The subtraction between the voltage max and the voltage at start counter divided by the data in time unit give us the value of the LSB.

In our case the LSB of the 12-bit ADC is 257μV.

Conversion formula:

$$\text{Analog voltage [mV]} = 970[\text{mV}] + \text{data} * \text{LSB}[\text{mV/adc}]$$

i. Pedestals 12-bit conversion

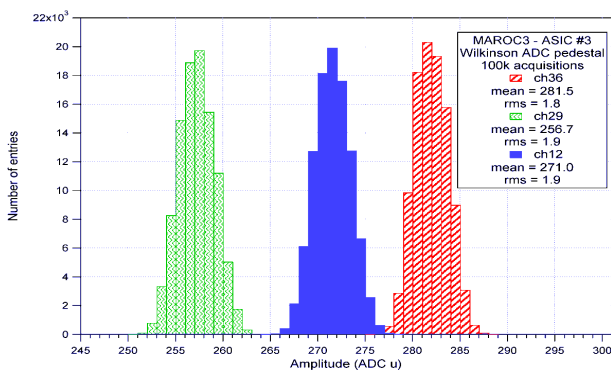


Figure 16: Histogramme of 3 channel pedestals via the 12-bit Wilkinson ADC

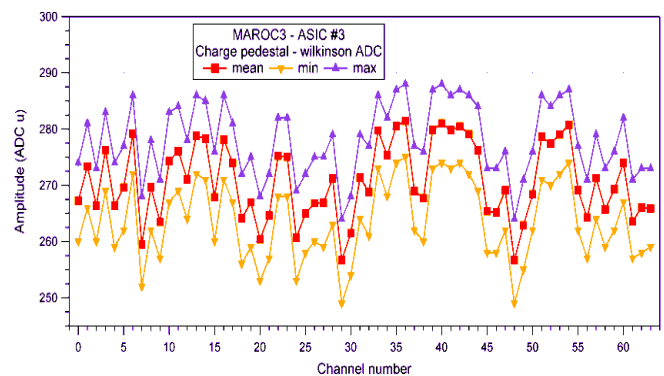


Figure 17: Mean and rms of charge pedestal via the ADC

The histogram rms corresponds of the noise of the slow shaper. In our case the slow shaper noise is 2 adc unit or about 514μV. The scope measurement gives about 580μV.

These graphs show a good uniformity for the pedestals and for the noise et the output of the slow shaper. The mean of the charge pedestal id about 270 adc unit or 1.039V.

➤ 10 bits and 8 bits conversion

10 bits	Tmax (μ s)	Vmax (V)	Data (adc unit)	Rms (adc unit)
0pC	4.12	1.06	60.5	0.55
0.5pC	6.11	1.15	143.4	0.56
1pC	7.82	1.23	215.1	0.55
10pC	19.12	1.75	662	0.52
50pC	21.50	1.86	756.6	0.53

8 bits	Tmax (μ s)	Vmax (V)	Data (adc unit)	Rms (adc unit)
0pC	1.05	1.06	14	0
0.5pC	1.54	1.16	33.5	0.5
1pC	1.97	1.24	50.5	0.5
10pC	4.64	1.76	157.6	0.48
50pC	5.22	1.87	180	0.03

In our case the LSB of the 10-bit ADC is 1.24mV and the LSB of the 8-bit ADC is 5.13mV.

➤ ADC conversion and readout Time

8.

	12bits	10bits	8bits
conversion	102 μ s	25 μ s	6 μ s
readout	19.2 μ s	16 μ s	12.8 μ s
Total	121.2 μ s	41 μ s	18.8 μ s
frequency	8 kHz	24kHz	53kHz

8. Fast channel

To deliver the trigger information there are 3 variable fast shapers and two discriminators. Two fast shapers (one unipolar and another bipolar) followed by one discriminator are dedicated for the photon electron counting and one bipolar fast shaper with lower gain followed by its discriminator has been added to provide a trigger signal at higher input charge. The 2 discriminator outputs are multiplexed to provide only 64 trigger outputs. The thresholds are loaded by 2 internal 10-bit DACs common for the 64 channels.

8.1. DAC

8.1.1. Principle

The threshold of each discriminator is set by a 10 bit-DAC which is different that those integrated in MAROC1 or MAROC2. The 2 DACs are identical except that the reference current for DAC0 can be chosen using SC parameter (small dac) equal or half of the Iref. This is to allow finest tunings of the thresholds for small dynamic range charge inputs. Reference voltages are referred to an integrated band gap ($v_{bg}=2.5V$).

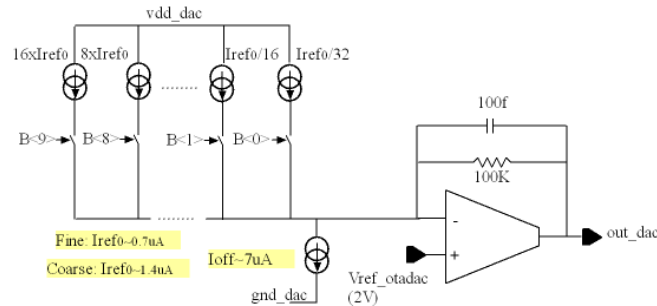


Figure 18: DAC schematic

8.1.2. DAC measurements

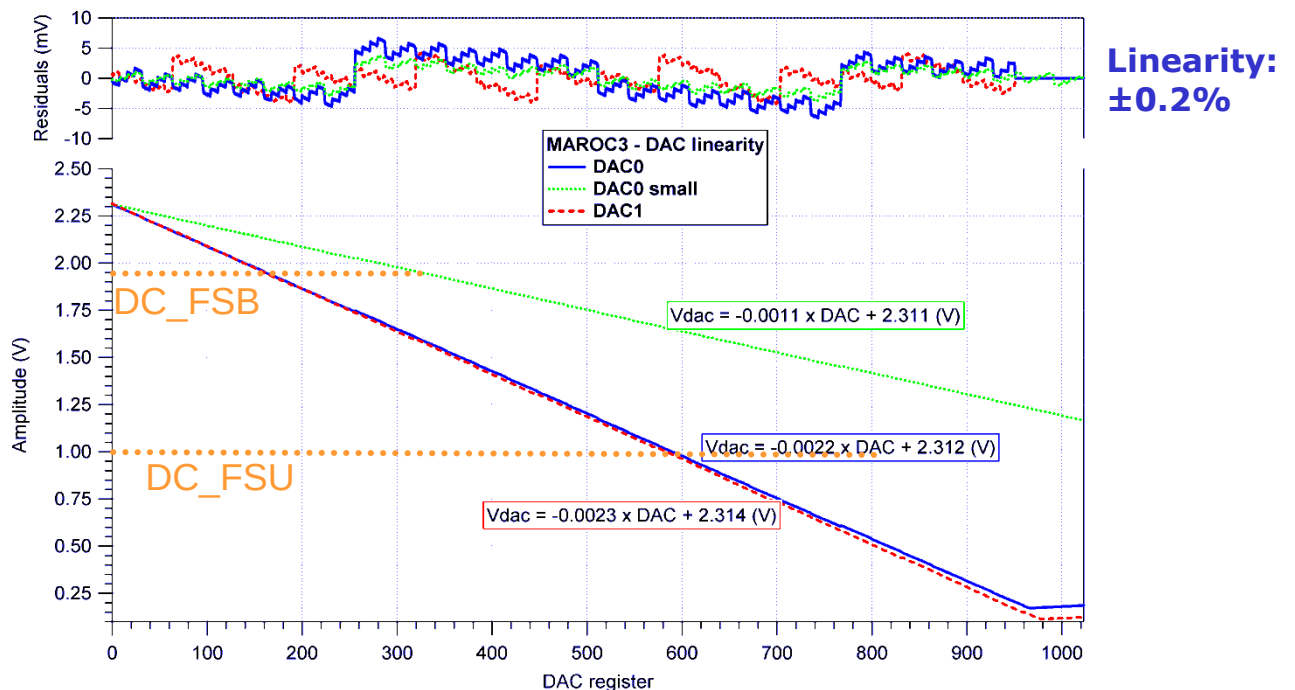


Figure 19: DAC linearity : simulation and measurement

	DAC0	DAC0 small	DAC1
slope	2.3mV/UDAC	1.1mV/UDAC	2.2mV/UDAC
DAC=0	2.3V	2.3V	2.3V

8.2. Variable Fast Shapers

There are 2 variable CRRC fast shapers. The network feedback of each amplifier can be changed independently thanks to the SC parameters. The peaking time is $\approx 20\text{-}25\text{ ns}$.

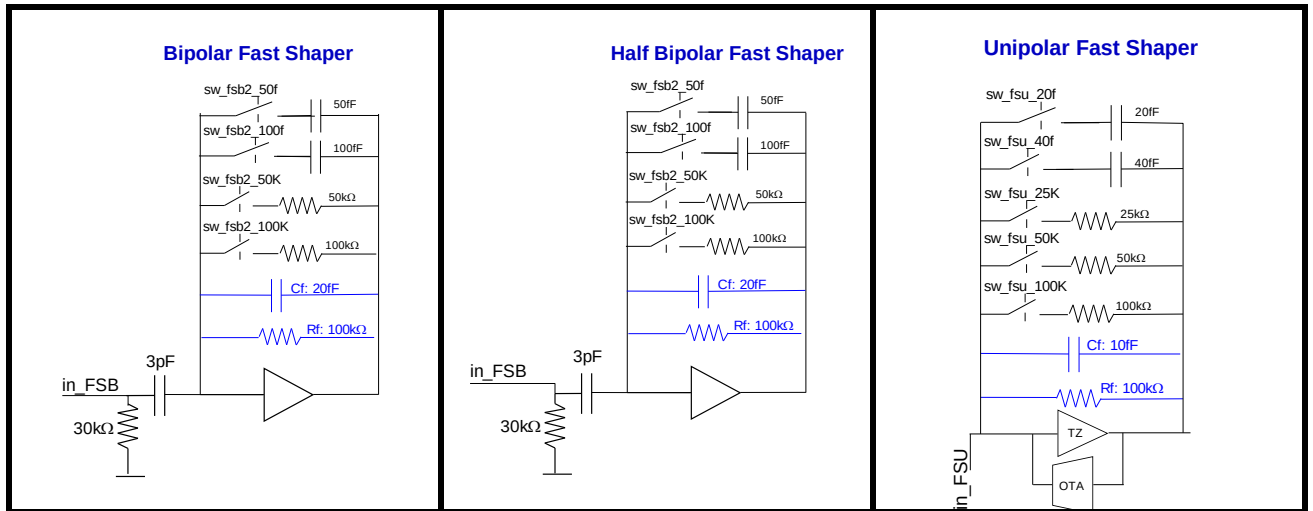


Figure 20: Fast Shapers schematics

8.3. Fast Shaper Measurements

8.3.1. DC measurements

The DC levels, named also pedestals, of each fast shaper outputs for all channels are measured.

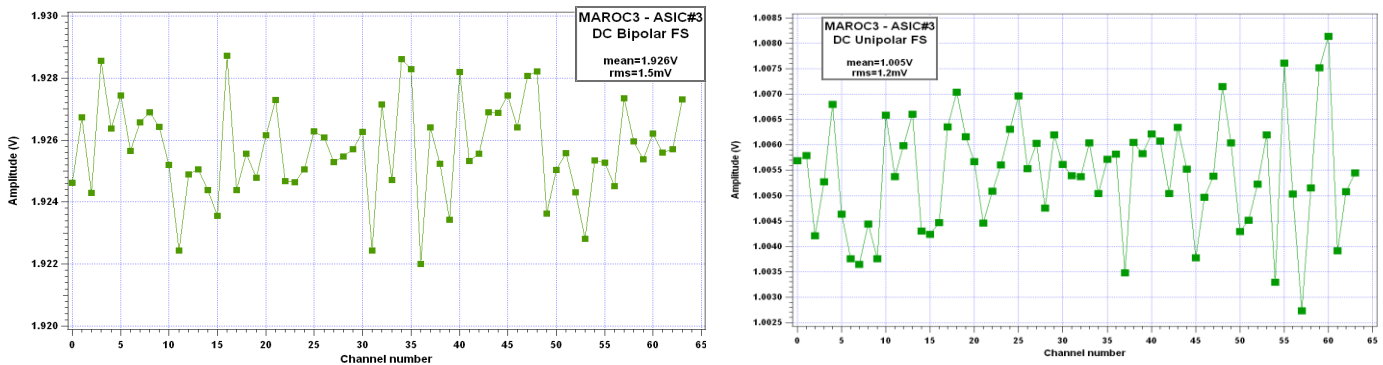


Figure 21: Pedestal uniformity: Bipolar and Unipolar DC level vs channel number

	Bipolar fast shaper n°1	Bipolar fast shaper n°2	Unipolar fast shaper
DC level	Mean=1.926V Rms=1.5mV	Mean=1.926V Rms=1.5mV	Mean=1.005V Rms=1.2mV
Equivalent DAC value	353 (small DAC)	169	569

9. Trigger outputs

9.1. S-curve vs injected charge

9.1.1. Bipolar Fast Shaper

In order to characterize the FSB1, the Scurves versus injected charge has been performed at variable DAC values (Figure. 22 left). All scurves have been fitted and charges corresponding of 50%trigger efficiency have been extracted and plotted versus the threshold.

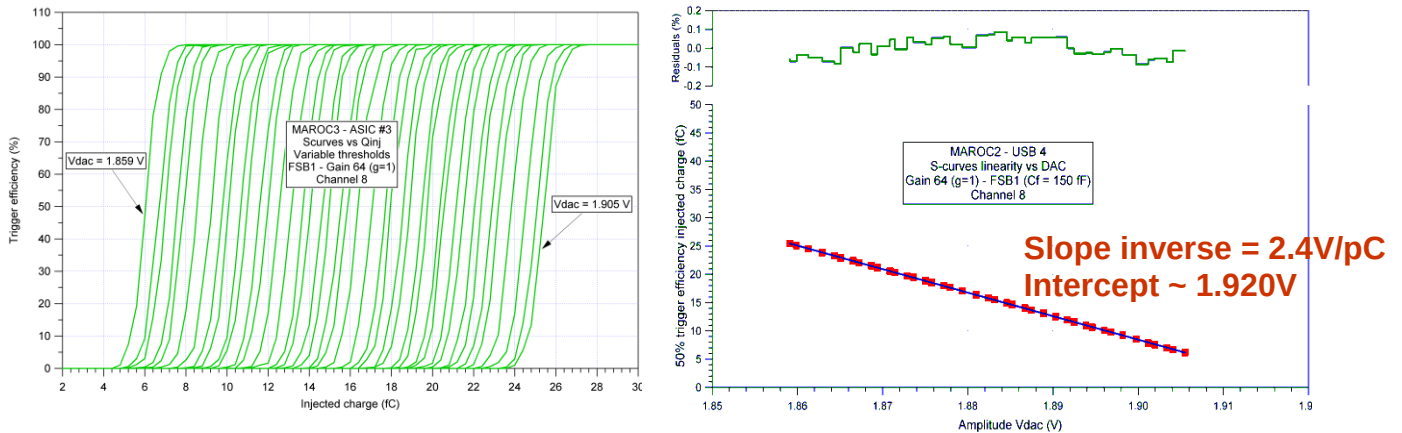


Figure 22: Left: Scurve vs Injected charge at different threshold, Right: 50%trigger efficiency versus threshold

The inverse slope is equivalent of the gain of the bipolar fast shaper: 2.4V/pC and the intercept is equivalent at the DC through the discriminator. This measure has been done for all channels to check the channel uniformity.

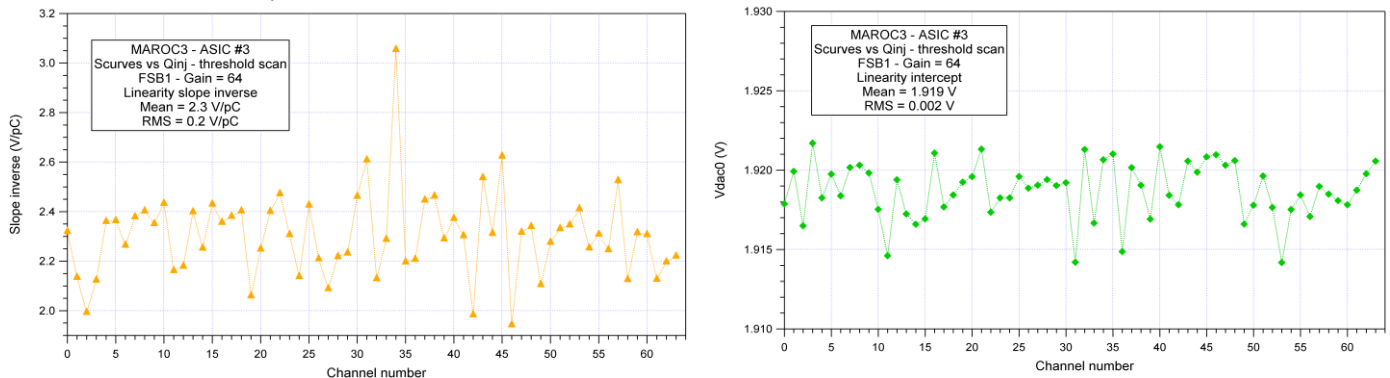


Figure 23: Left: FSB1 gain for all channels, Right: intercept of the fit

9.1.2. Unipolar Fast Shaper

Same measurements have been performed with the unipolar fast shaper.

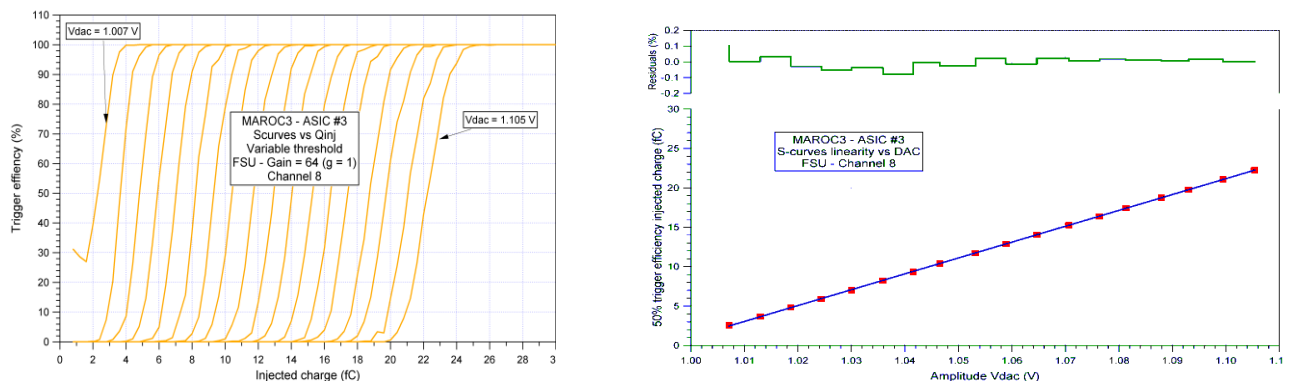


Figure 24: Left: Scurve vs injected charge at different threshold, Right: 50%trigger efficiency versus threshold

The inverse of the slope gives the Unipolar fast shaper gain: 4.4V/pC

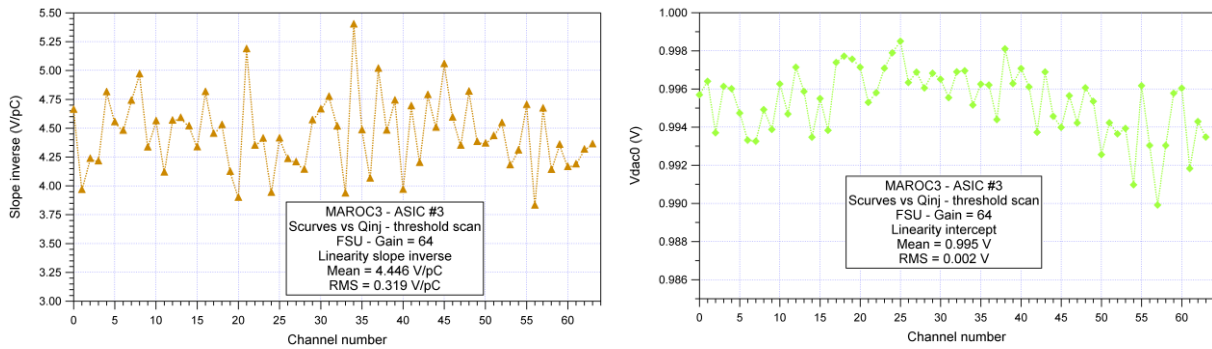


Figure 25: Left: FSU gain for all channels , Right: intercept of the fit

Very good results in trigger measurements have been obtained. Thanks to the S-Curve tests with different injected charges, it has been demonstrated that the discriminator threshold can be set at lower than 1/3 of pe for the bipolar and the unipolar fast shaper.

9.2. S-curve vs injected charge at different feedback resistors settings

9.2.1. Bipolar Fast Shaper

The FS feedback resistor is made of one 100kΩ resistor and in parallel of two resistors (50kΩ and 100kΩ) can be added by slow control register. The feedback resistors chosen modify the fast shaper gain.

Rf settings	Equivalent resistor	Gain (V/pC)	Minimum charge detected
0kΩ	100kΩ	2.32	4fC
100kΩ	50kΩ	1.89	7.5fC
50kΩ	33kΩ	1.56	10fC
150kΩ	25kΩ	1.32	

9.2.2. Gain adjustment

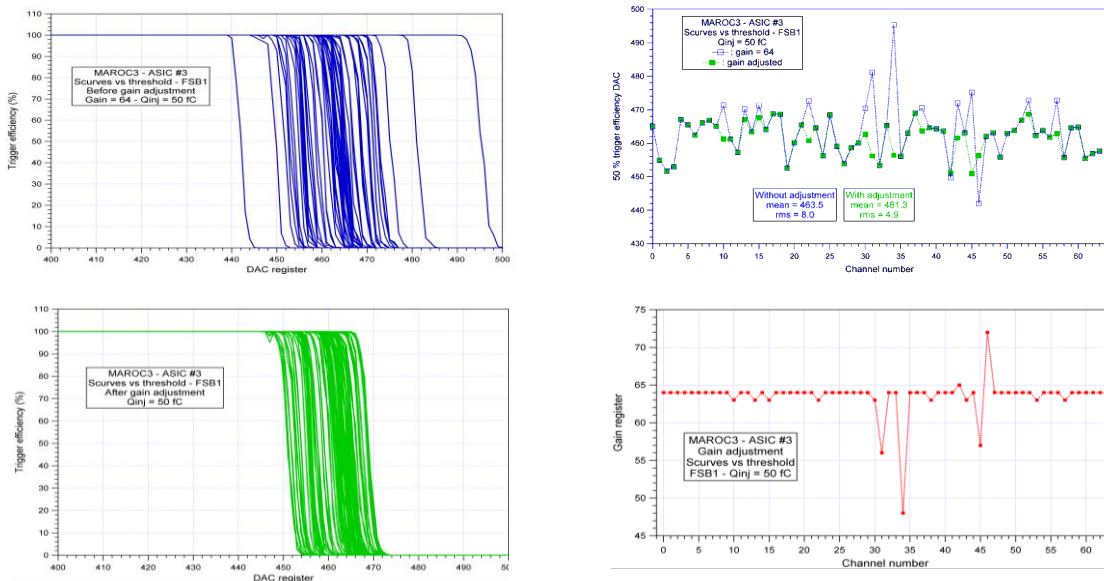


Figure 26: Preamplifier gain adjustment for the bipolar fast shaper

The gain adjustment improve the trigger uniformity inside the asic.

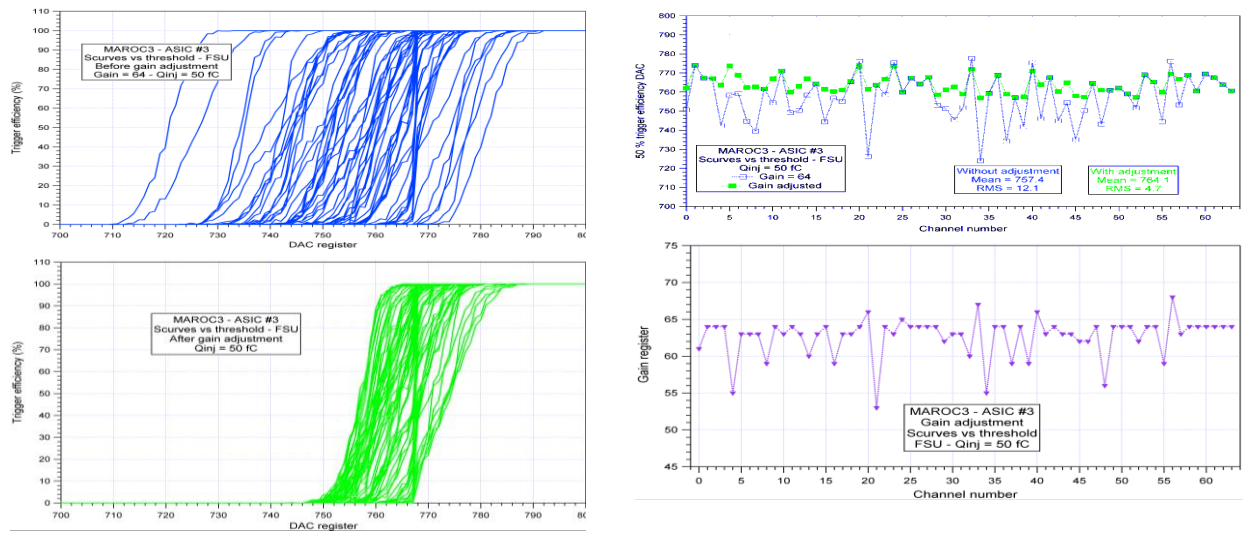
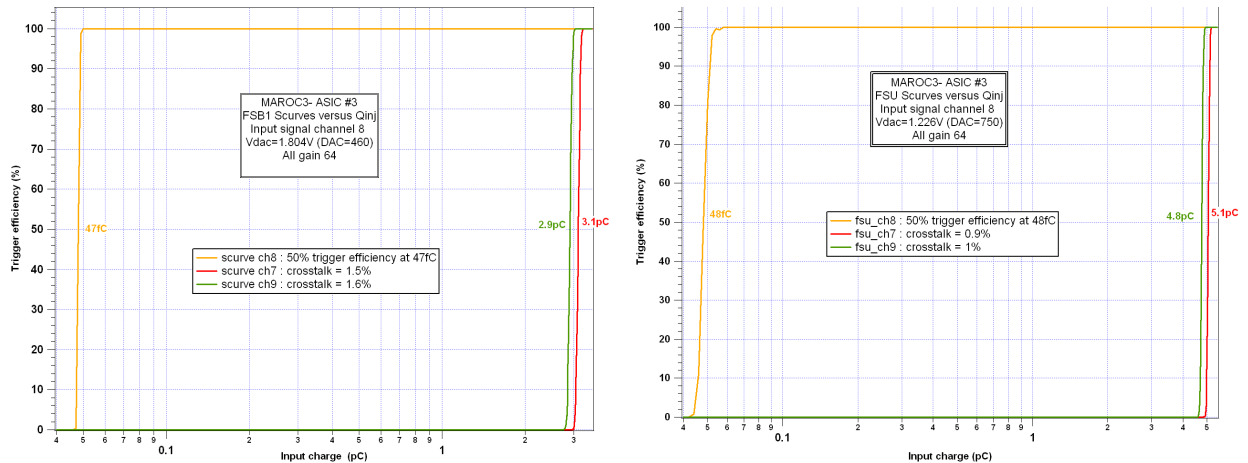


Figure 27: Preamplifier gain adjustment for the unipolar fast shaper

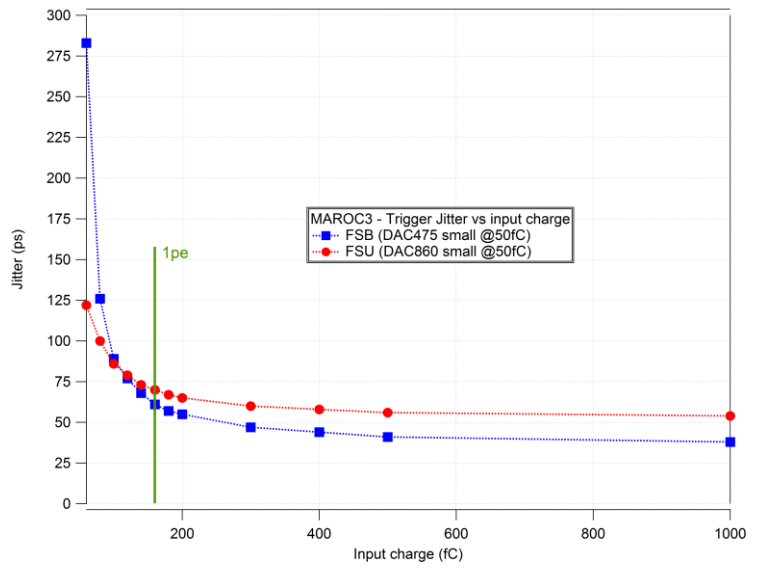
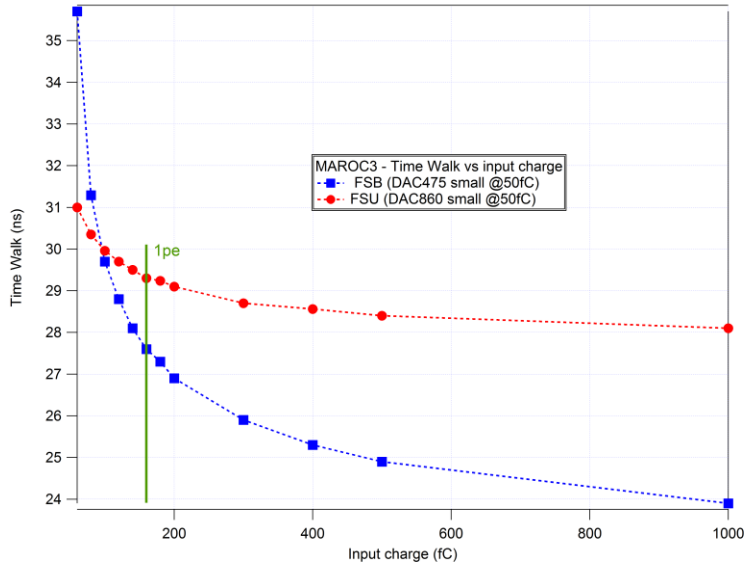
9.3. Cross talk



FSB	50% trigger efficiency	Cross talk
ch8	47fC	
ch7	3.1pC	1.5%
ch9	2.9pC	1.6%

FSU	50% trigger efficiency	Cross talk
ch8	48fC	
ch7	5.1pC	0.9%
ch9	4.8pC	1%

9.1.Jitter et Time walk measurements



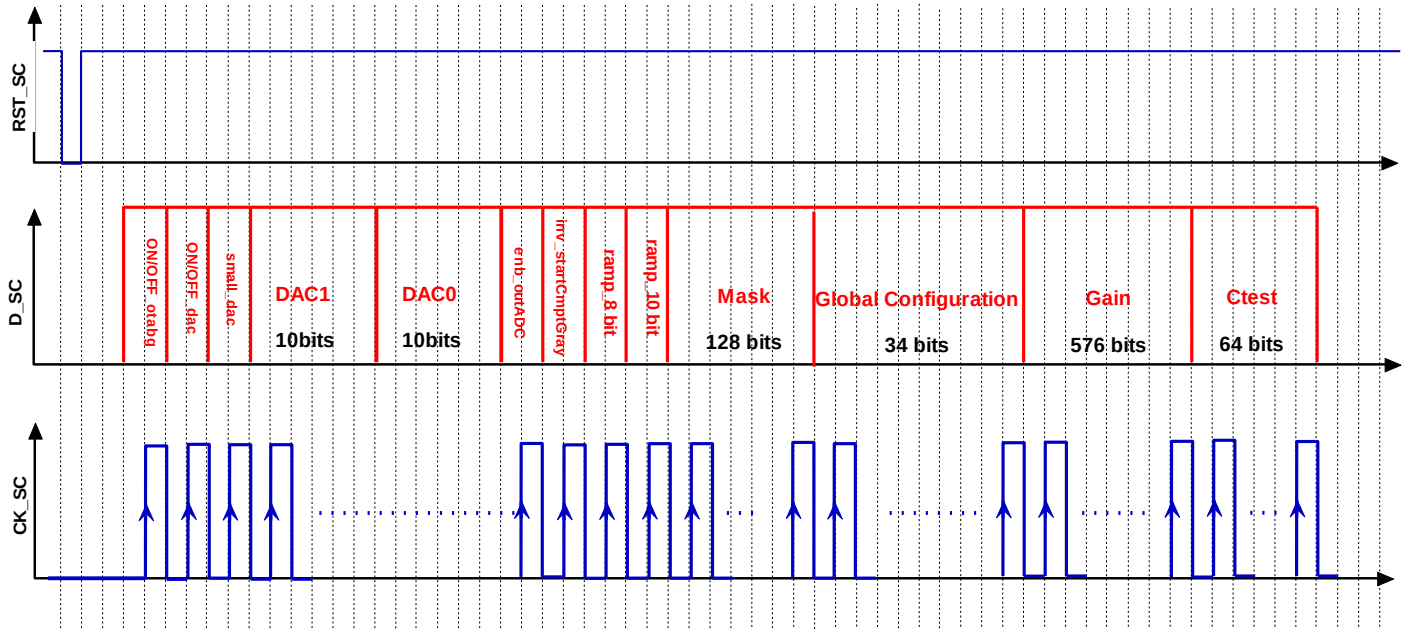
10. Result summary

	MAROC3	
	consumption	220mW (→ 3.5mW/channel)
inputs	number	64
	polarity	negative
	Input impedance	about 50-60 Ω
outputs	triggers	64 triggers (Vhi=1.5V and Vlo=0V)
	charge	1 analog multiplex output 1 digital charge (12, 10 or 8 bits)
PA	Gain variable	8 bits (0 to ~4)
BFS	Gain	4.5V/pC
	Noise	1.6mV
	Min charge	5fC
	Jitter@160fC (for DAC@50fC)	61ps
UFS	Gain	2.3V/pC
	Noise	2.4mV
	Min charge	3fC
	Jitter@160fC (for DAC@50fC)	70ps
Slow shaper	Gain	55mV/pC (gain64) – shaping=160ns
	Noise	0.58mV

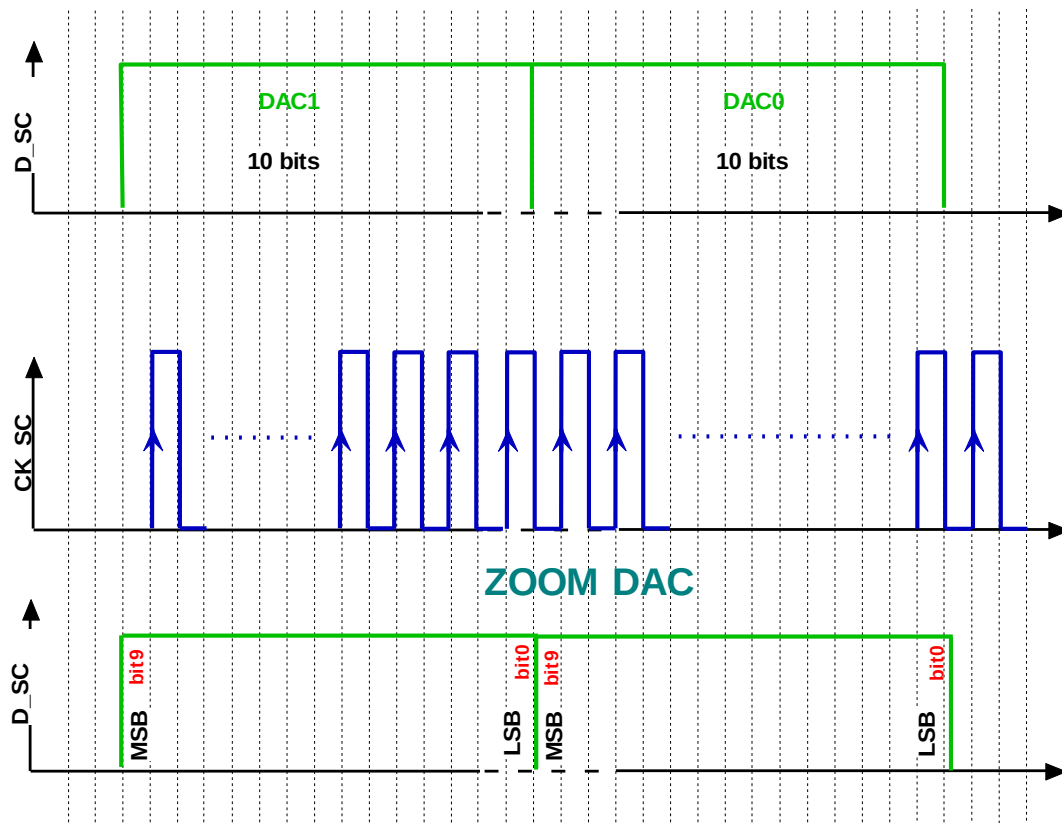
11. TEST BOARD

11.1. ASIC Registers

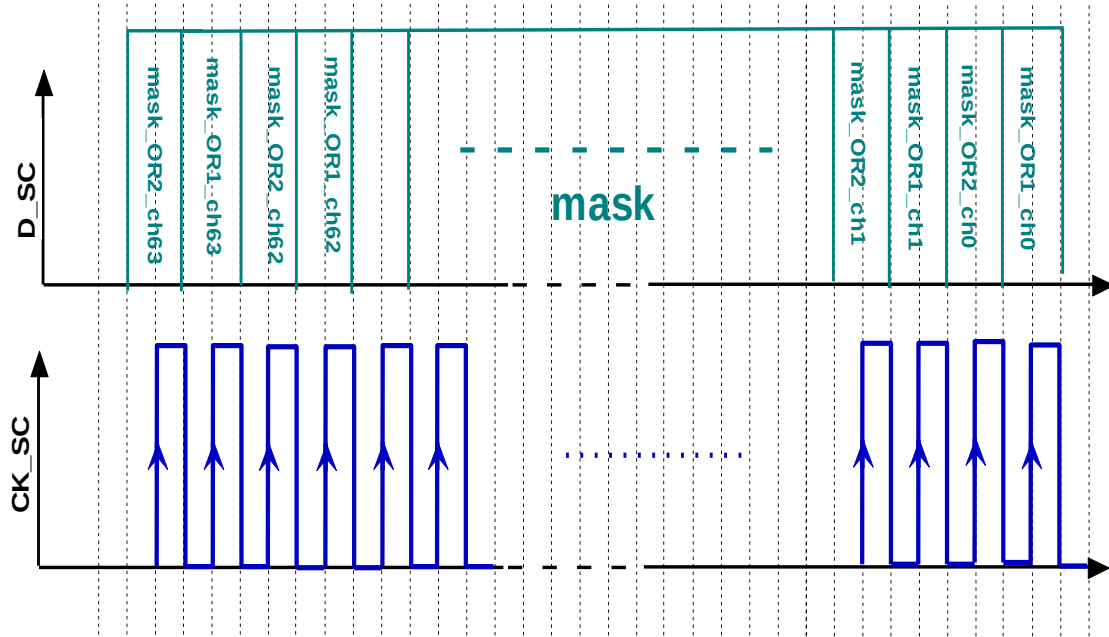
11.1.1. D_SC, CK_SC and RST_SC (Static Register)



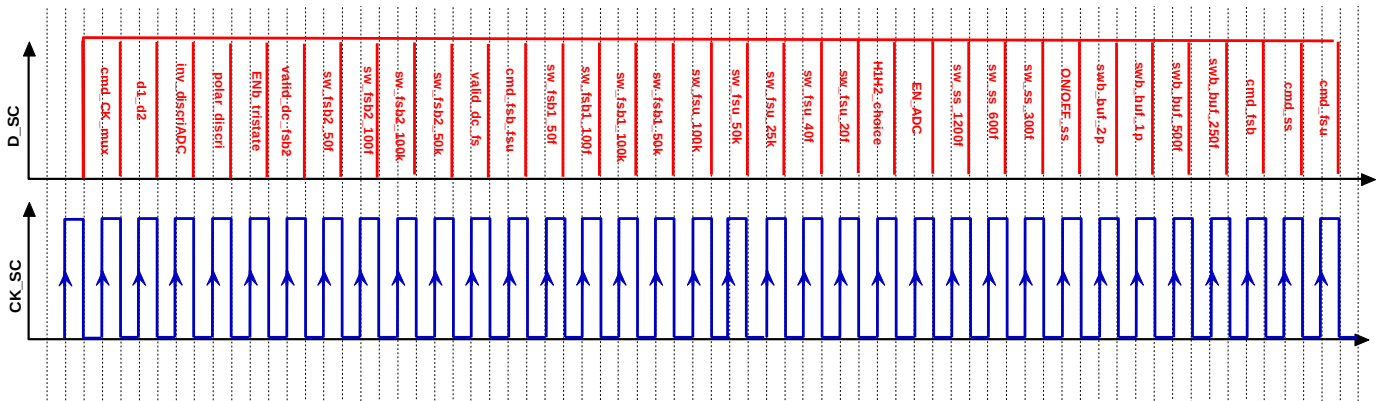
DACs bits



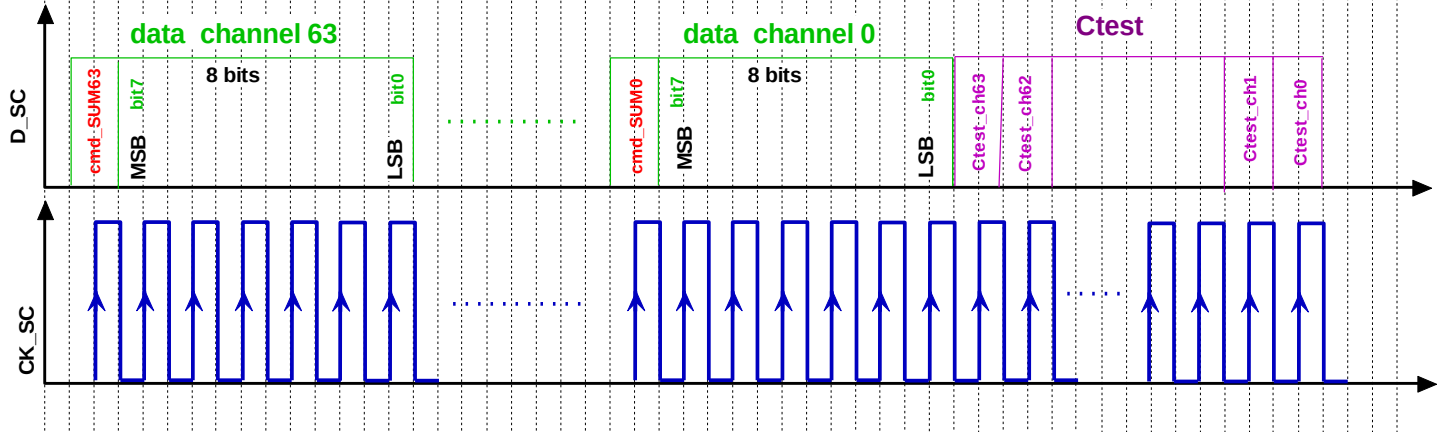
Mask bits



Global Config



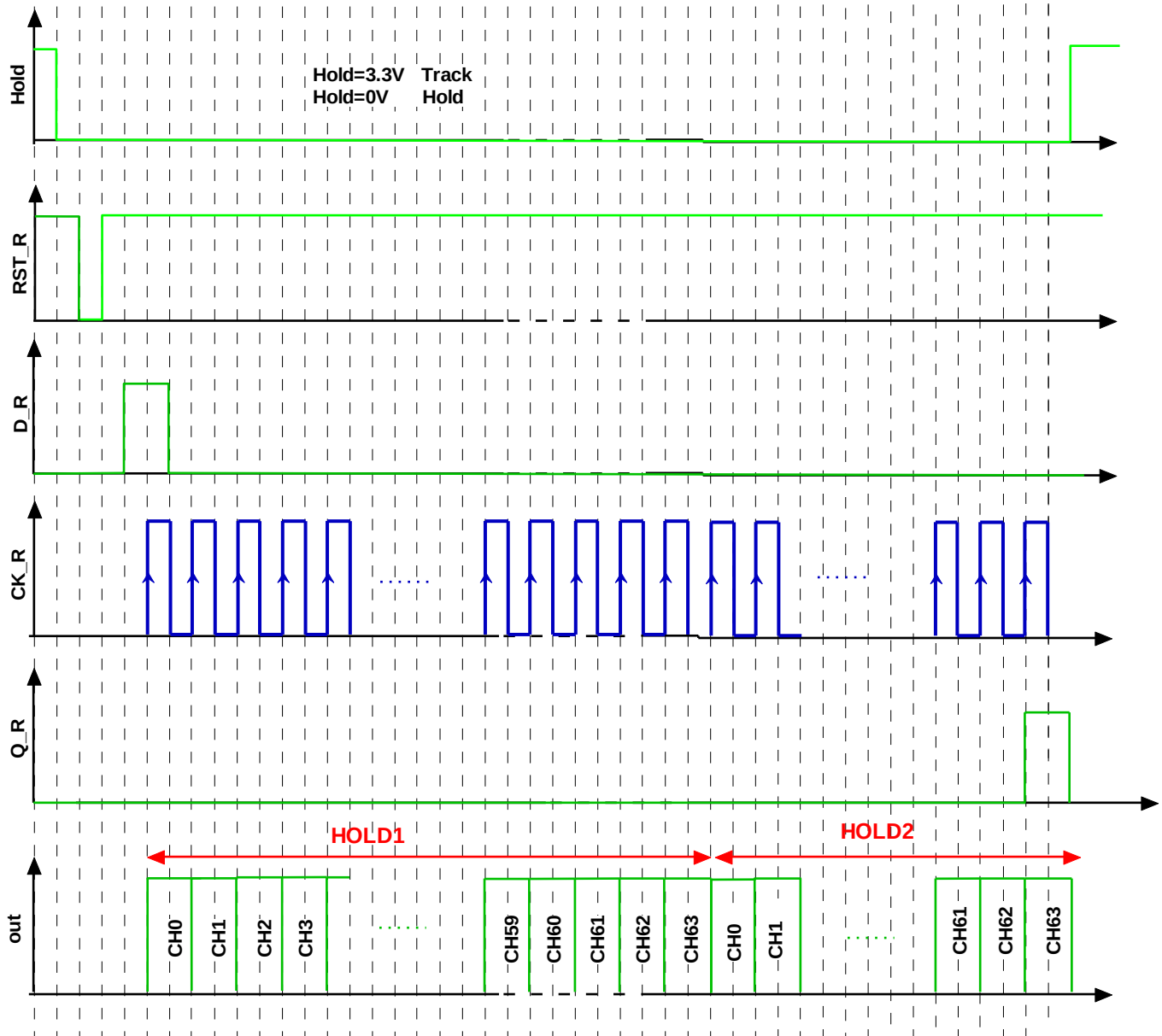
Gain, Sum and Ctest



11.1.2. D_R, CK_R and RST_R* (Dynamic Register)

➤ For multiplex charge output

Multiplex charge output



➤ For DC_FS output (pin 201)

SC bits and name		description	observations
0	ON/OFF_otabg	power pulsing bit for bandgap	not active on MAROC3 test board because power pulsing pin is connected to vdd
1	ON/OFF_dac	power pulsing bit for all DACs	not active on MAROC3 test board because power pulsing pin is connected to vdd
2	small_dac	to decrease the slope of DAC0 -> better accuracy	small dac OFF: threshold VTH0 min= and max= small dac ON: threshold VTH0 min= and max=
3	DAC1[9]	DAC value for the second discr (with the fast shaper FSB2)	
4	DAC1[8]		
	'''		
11	DAC1[1]		
12	DAC1[0]		
13	DAC0[9]	DAC value for the first discr (with the fast shaper FSB1 or FSU)	
14	DAC0[8]		
	'''		
21	DAC0[1]		
22	DAC0[0]		
23	enb_outADC	wilkinson ADC parameter: enable data output	to use the wilkinson ADC should be OFF
24	inv_startCmptGray	wilkinson ADC parameter: the start compteur polarity switching	to use the wilkinson ADC should be OFF
25	ramp_8bit	wilkinson ADC parameter: ramp slope change to have quickly conversion on 8 bits	
26	ramp_10bit	wilkinson ADC parameter: ramp slope change to have quickly conversion on 10 bits	
27	mask_OR2_ch63	mask the second discr output of ch63 (FSB2 to generate the trigger)	mask bit ON: no trigger output , mask bit OFF: trigger output visible
28	mask_OR1_ch63	mask the first discr output of ch63 (FSB1 or FSU to generate the trigger)	mask bit ON: no trigger output , mask bit OFF: trigger output visible
29	mask_OR2_ch62	mask the second discr output of ch62 (FSB2 to generate the trigger)	
30	mask_OR1_ch62	mask the first discr output of ch62 (FSB1 or FSU to generate the trigger)	
	'''		
151	mask_OR2_ch1	mask the second discr output of ch1 (FSB2 to generate the trigger)	
152	mask_OR1_ch1	mask the first discr output of ch1 (FSB1 or FSU to generate the trigger)	
153	mask_OR2_ch0	mask the second discr output of ch0 (FSB2 to generate the trigger)	
154	mask_OR1_ch0	mask the first discr output of ch0 (FSB1 or FSU to generate the trigger)	
155	cmd_CK_mux		Should be OFF
156	d1_d2	trigger output choice	d1_d2='0' -> trigger from FSB1 and DAC0 ; d1_d2='1' -> trigger from FSB2 and DAC1
157	inv_discrADC	ADC discr output could be inverted	Should be OFF
158	polar_discr	polarity of trigger output	polar_discr='0' ->High polarity ; polar_discr='1' -> Low polarity
159	Enb_tristate	enable all trigger outputs	Should be ON to see trigger outputs
160	valid_dc_fsb2	enable FSB2 DC measurements	
161	sw_fsb2_50f	Feedback capacitor for FSB2	better if ON
162	sw_fsb2_100f	Feedback resistor for FSB2	
163	sw_fsb2_100k	Feedback resistor for FSB2	
164	sw_fsb2_50k	Feedback resistor for FSB2	
165	valid_dc_fs	enable FSB and FSU DC measurements	
166	cmd_fsb_fsu	Choice between FSB1 or FSU for the first discr input (with DAC0)	cmd_fsb_fsu='1'-> FSU ; cmd_fsb_fsu='0'-> FSB

167	sw_fsb1_50f	Feedback capacitor for FSB1	better if ON
168	sw_fsb1_100f	Feedback capacitor for FSB1	better if ON
169	sw_fsb1_100k	Feedback resistor for FSB1	
170	sw_fsb1_50k	Feedback resistor for FSB1	
171	sw_fsu_100k	Feedback resistor for FSU	
172	sw_fsu_50k	Feedback resistor for FSU	
173	sw_fsu_25k	Feedback resistor for FSU	
174	sw_fsu_40f	Feedback capacitor for FSU	better if ON
175	sw_fsu_20f	Feedback capacitor for FSU	better if ON
176	H1H2_choice	ADC wilkinson: choice between the first or the second track and hold for the input of the ADC	
177	EN_ADC	ADC wilkinson: enable ADC conversion inside the asic	Should be ON to make a conversion
178	sw_ss_1200f	Feedback capacitor for Slow Shaper	
179	sw_ss_600f	Feedback capacitor for Slow Shaper	
180	sw_ss_300f	Feedback capacitor for Slow Shaper	
181	ON/OFF_ss	Power supply of Slow Shaper	
182	swb_buf_2p	capacitor for the buffer before the slow shaper	
183	swb_buf_1p	capacitor for the buffer before the slow shaper	
184	swb_buf_500f	capacitor for the buffer before the slow shaper	
185	swb_buf_250f	capacitor for the buffer before the slow shaper	
186	cmd_fsb	enable signal at the FSB inputs	Should be ON if we want to use FSB1 or FSB2
187	cmd_ss	enable signal at the SS inputs	Should be ON if we want to do charge measurement
188	cmd_fsu	enable signal at the FSU inputs	Should be ON if we want to use FSU
189	cmd_SUM63	enable signal to do sum	
190	GAIN63[7]	preamplifier gain value channel 63	
191	GAIN63[6]		
192	GAIN63[5]		
193	GAIN63[4]		
194	GAIN63[3]		
195	GAIN63[2]		
196	GAIN63[1]		
197	GAIN63[0]		
198	cmd_SUM62	enable signal to do sum	
199	GAIN62[7]	preamplifier gain value channel 63	
200	GAIN62[6]		
201	GAIN62[5]		
202	GAIN62[4]		
203	GAIN62[3]		
204	GAIN62[2]		
205	GAIN62[1]		
206	GAIN62[0]		
	///		
759	GAIN0[5]	preamplifier gain value channel 0	
760	GAIN0[4]		
761	GAIN0[3]		
762	GAIN0[2]		
763	GAIN0[1]		
764	GAIN0[0]		
765	Ctest_ch63	enable signal in Ctest input	
766	Ctest_ch62	enable signal in Ctest input	

767	Ctest_ch61	enable signal in Ctest input	
	'''	enable signal in Ctest input	
826	Ctest_ch2	enable signal in Ctest input	
827	Ctest_ch1	enable signal in Ctest input	
828	Ctest_ch0	enable signal in Ctest input	